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(54) **HIGH-SPEED READOUT IMAGE SENSOR**

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(57) **ABSTRACT**

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Related U.S. Application Data

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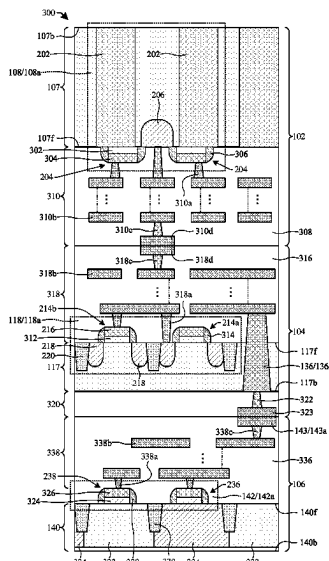
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H10F 39/00 (2025.01)

(52) **U.S. Cl.**
CPC **H10F 39/811** (2025.01); **H10F 39/809** (2025.01); **H01L 2225/06544** (2013.01)

(58) **Field of Classification Search**
CPC H10F 39/811; H10F 39/809
See application file for complete search history.

Various embodiments of the present disclosure are directed towards an image sensor. The image sensor includes a first chip bonded to a second chip. The first chip includes a semiconductor substrate. The first chip includes a first transistor cell and a second transistor cell. The second transistor cell is laterally spaced from the first transistor cell. A first through-substrate via (TSV) extends vertically through the semiconductor substrate. The first transistor cell is electrically coupled to the first TSV. A second TSV extends vertically through the first semiconductor substrate. The second transistor cell is electrically coupled to the second TSV. The second chip comprises a first readout circuit that is electrically coupled to the first TSV and the second TSV. The first readout circuit is disposed laterally between the first TSV and the second TSV. The first readout circuit is configured to receive a first signal from the first transistor cell.

20 Claims, 20 Drawing Sheets



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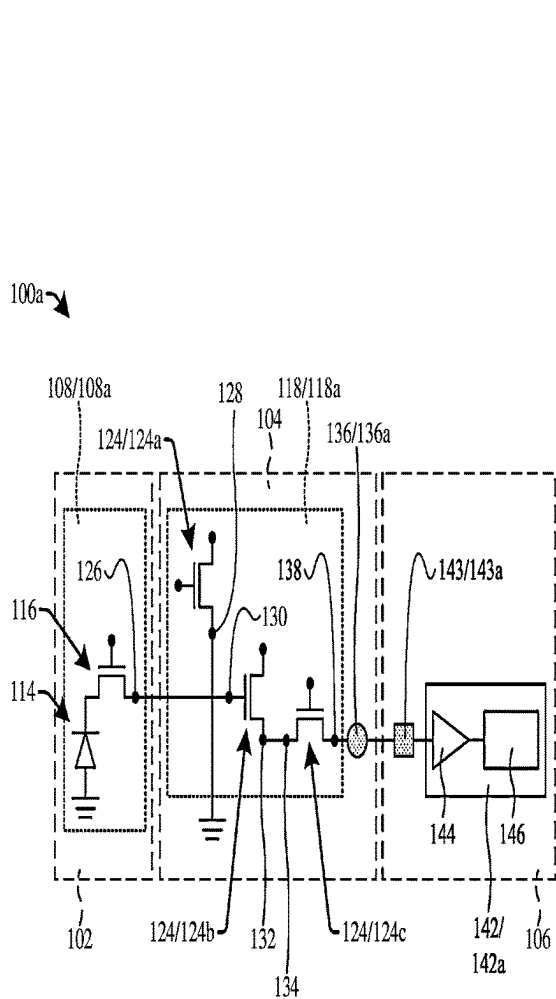


Fig. 1A

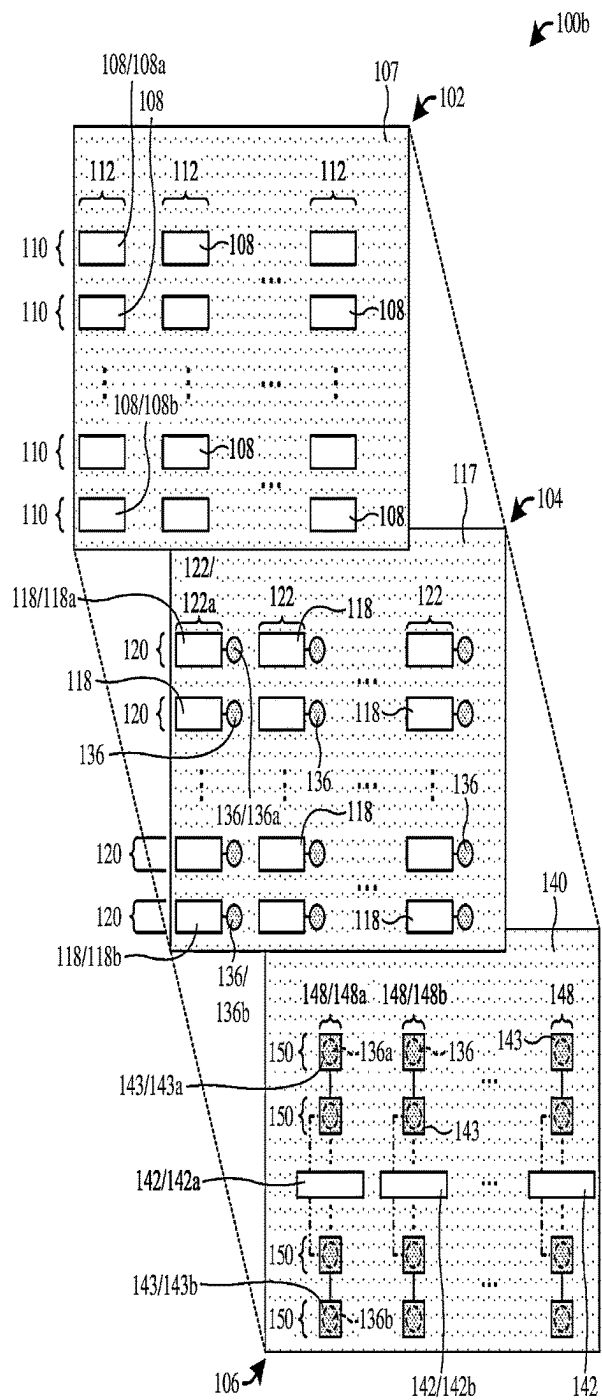


Fig. 1B

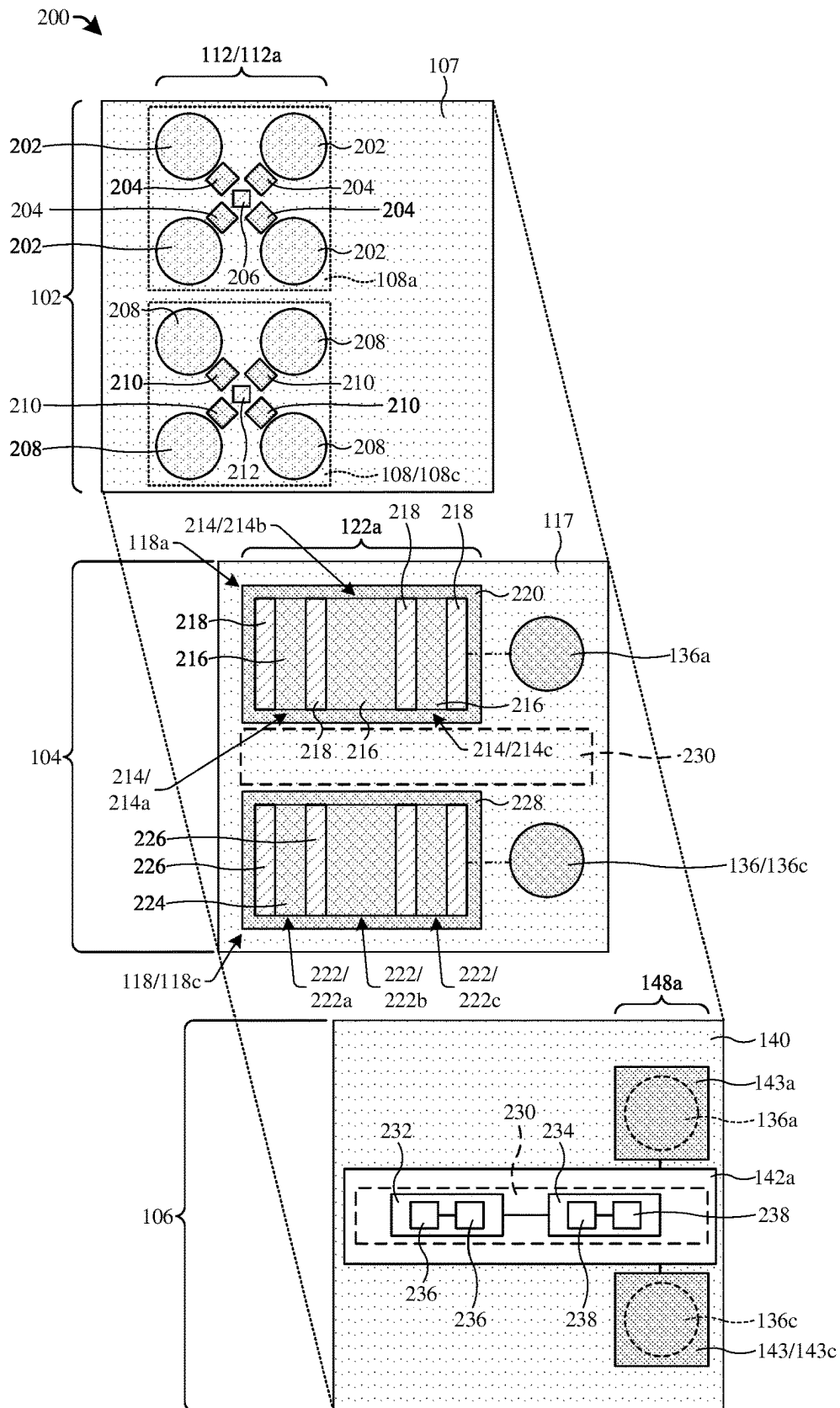


Fig. 2

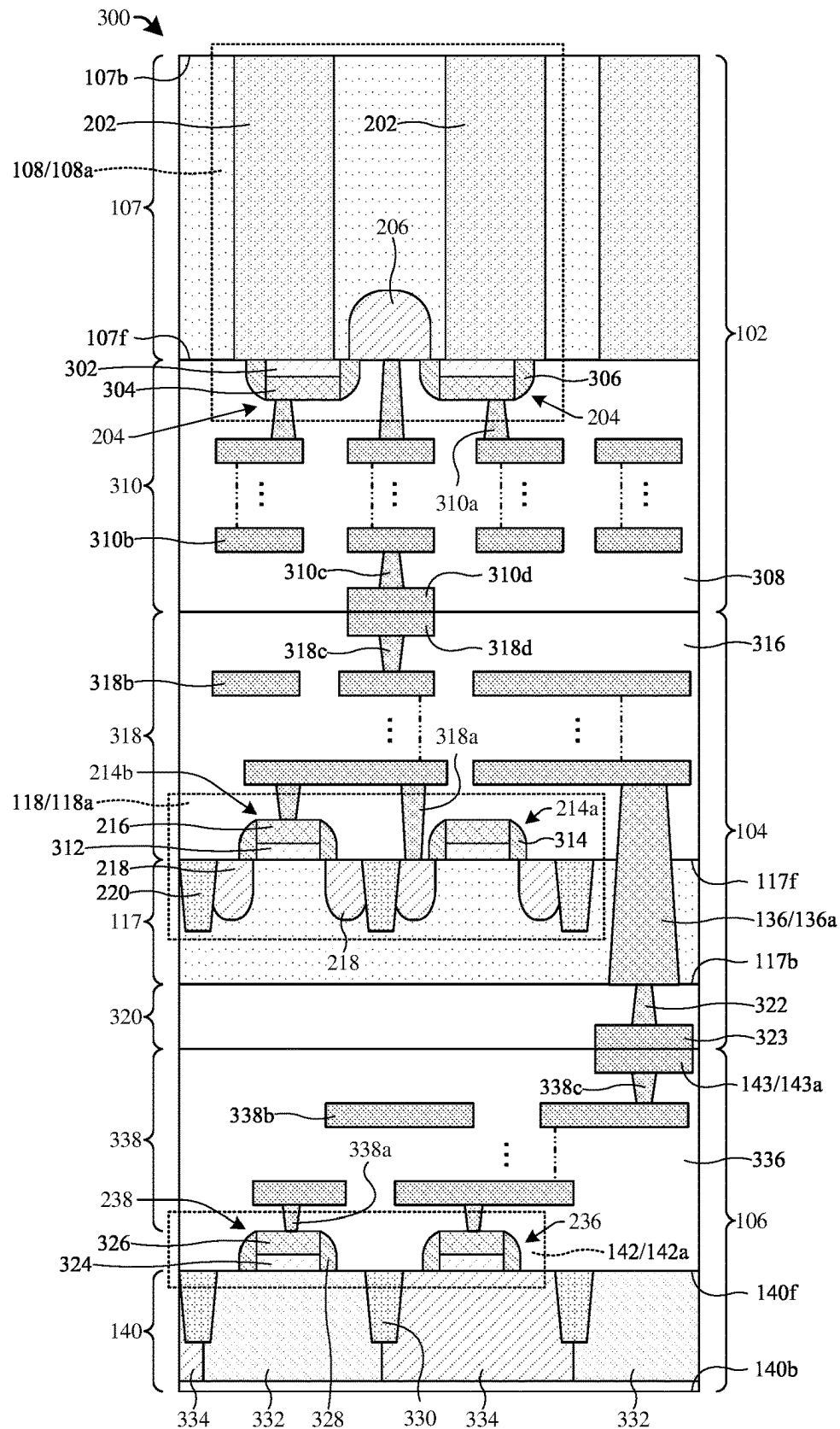


Fig. 3

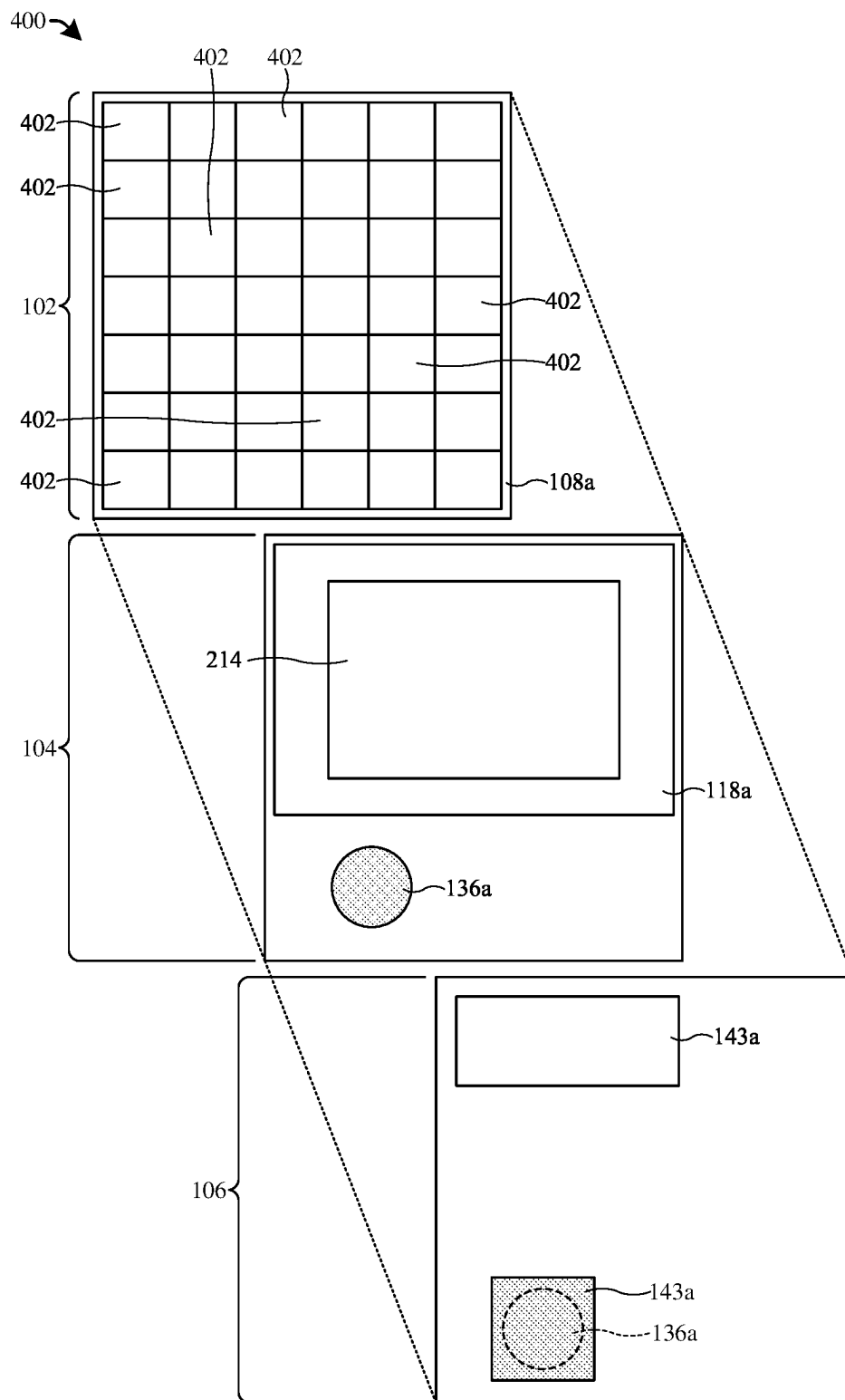


Fig. 4

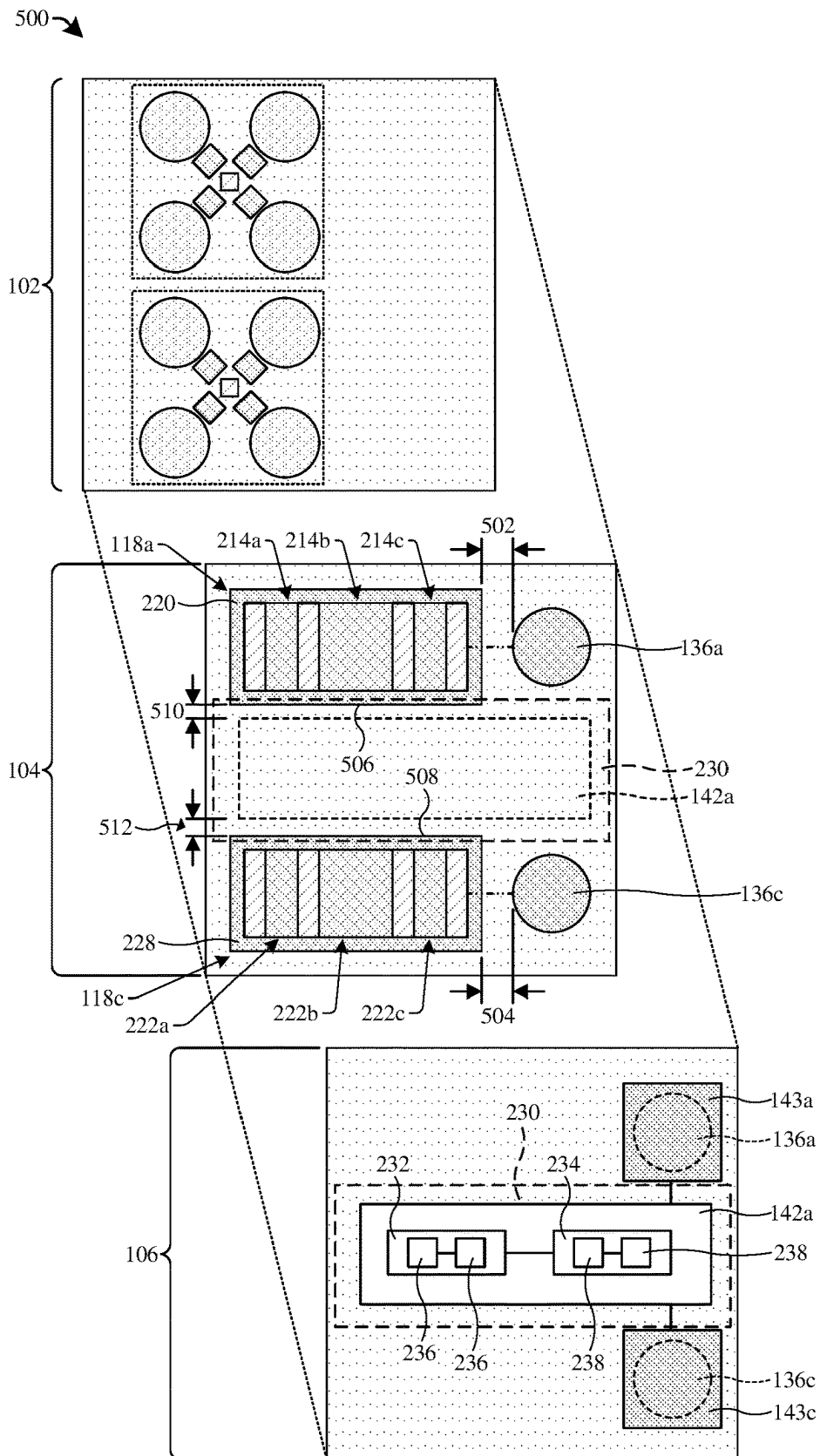


Fig. 5

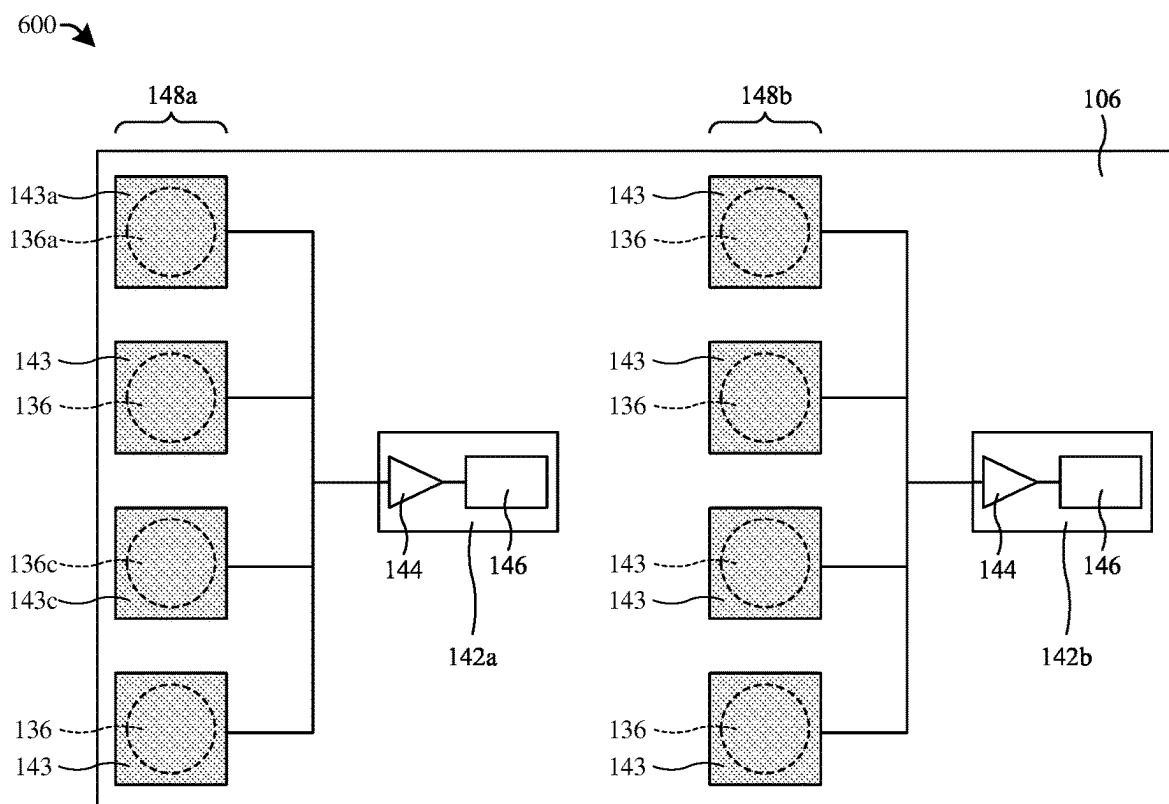


Fig. 6

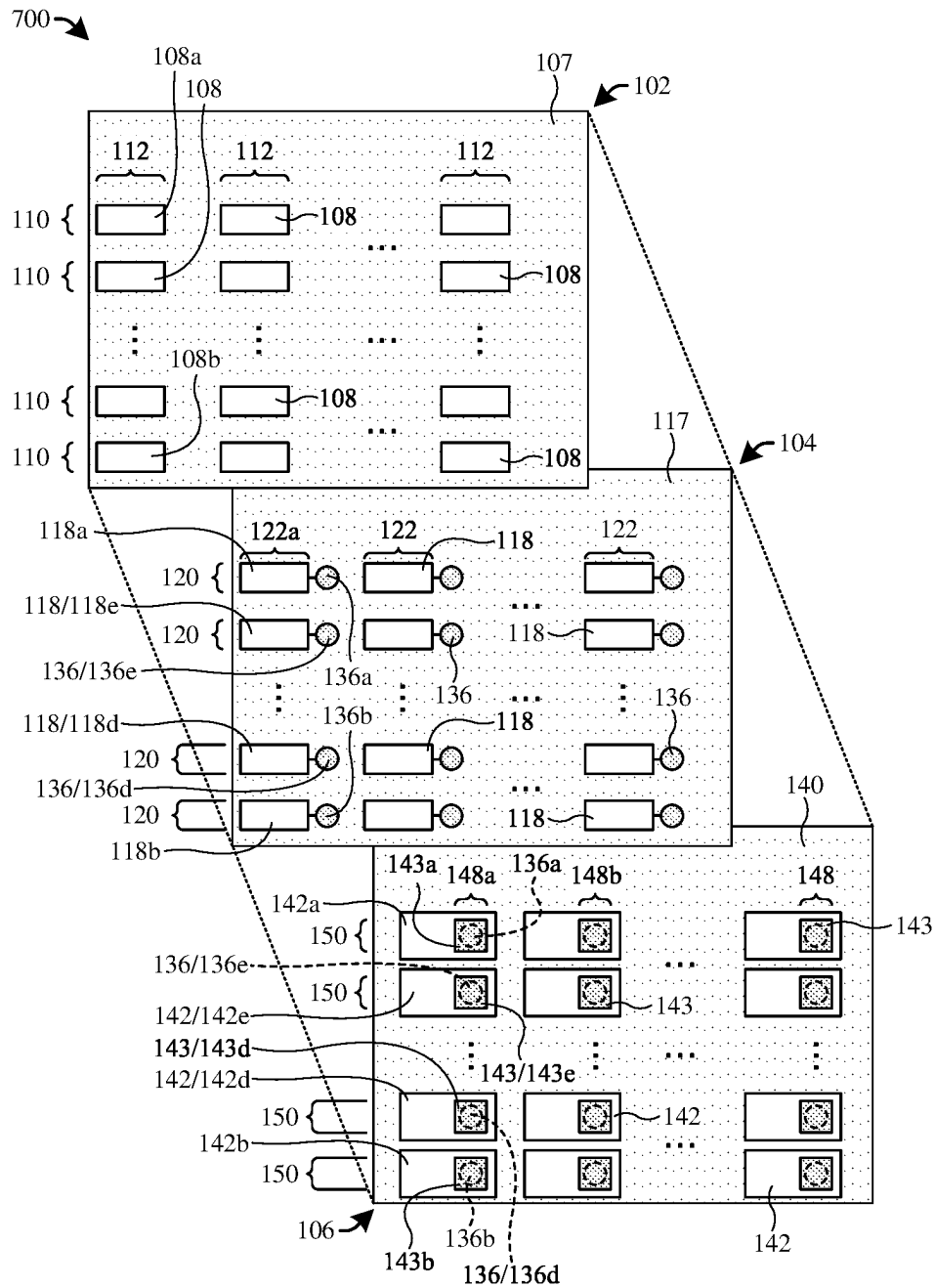
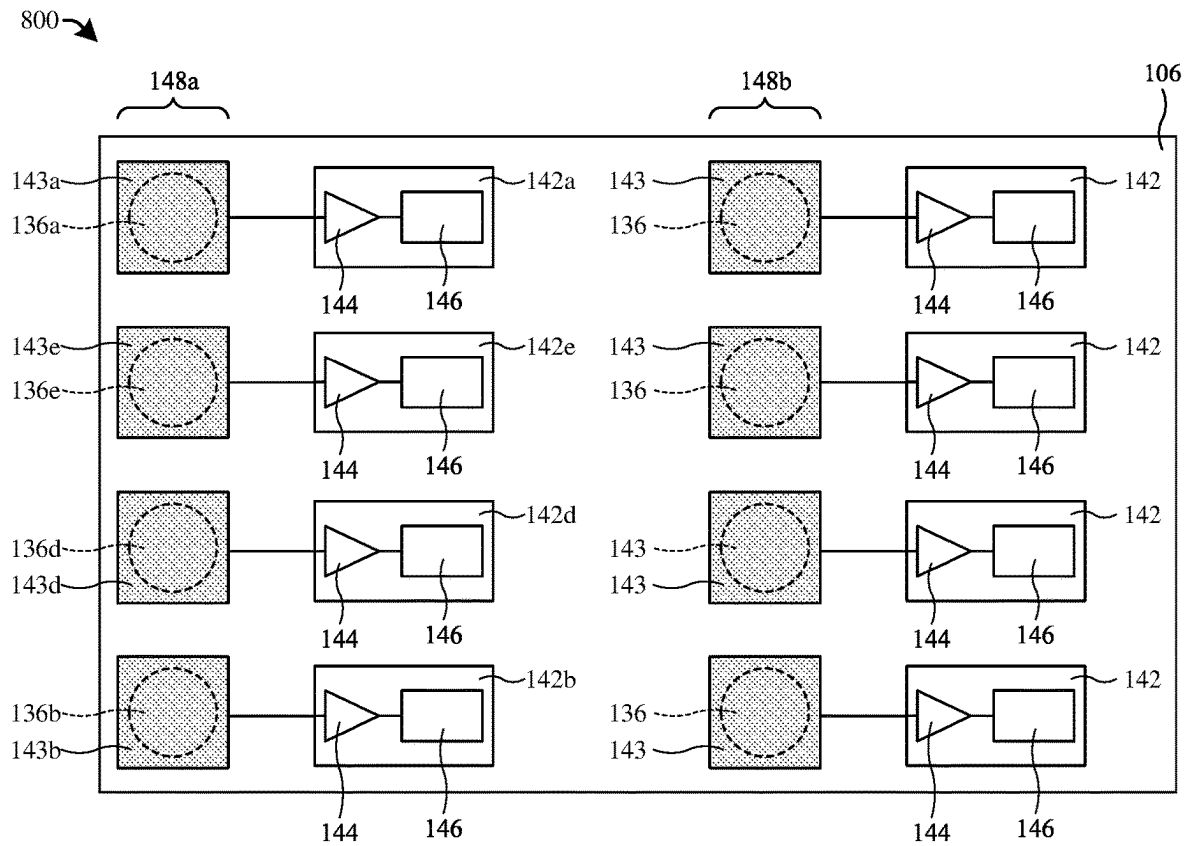


Fig. 7

**Fig. 8**

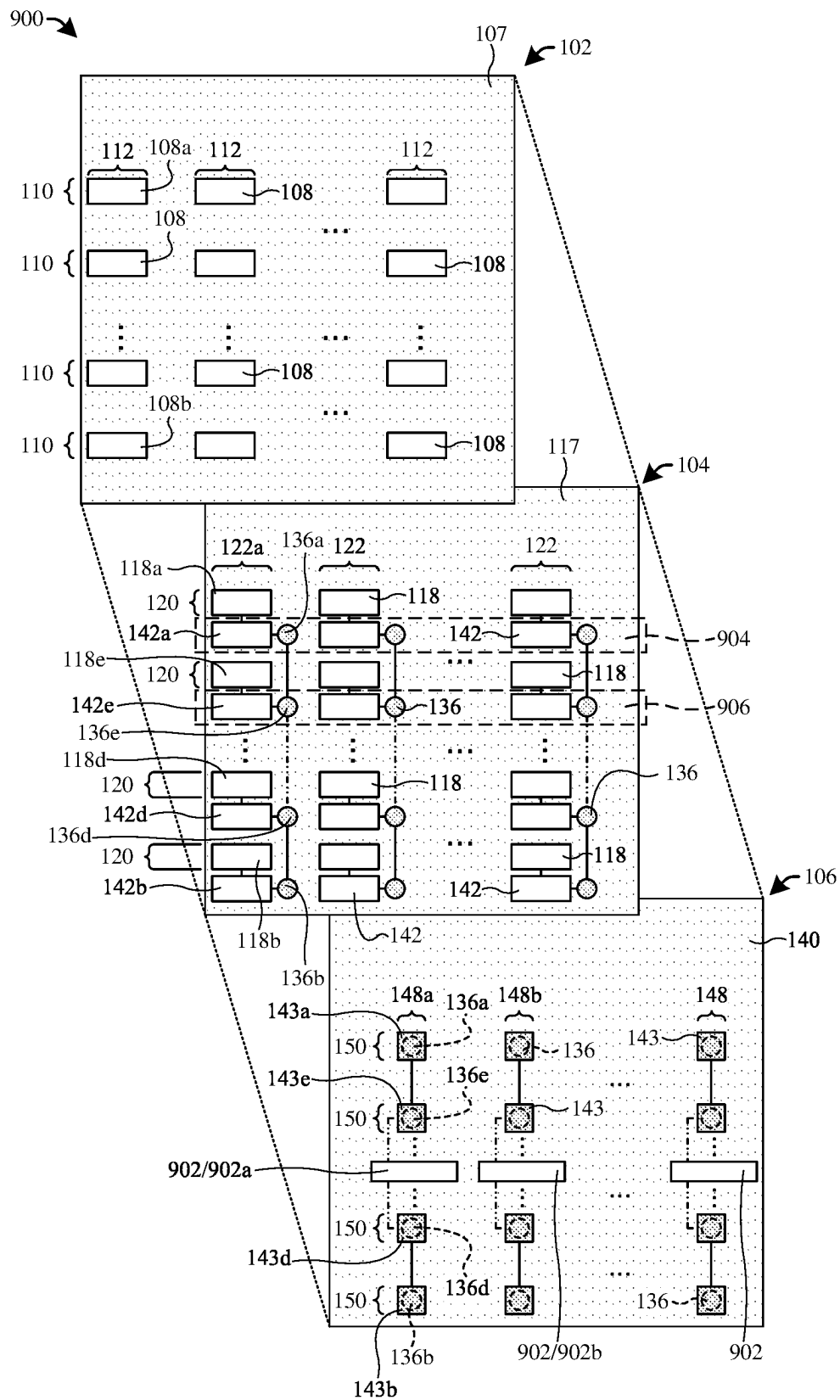


Fig. 9

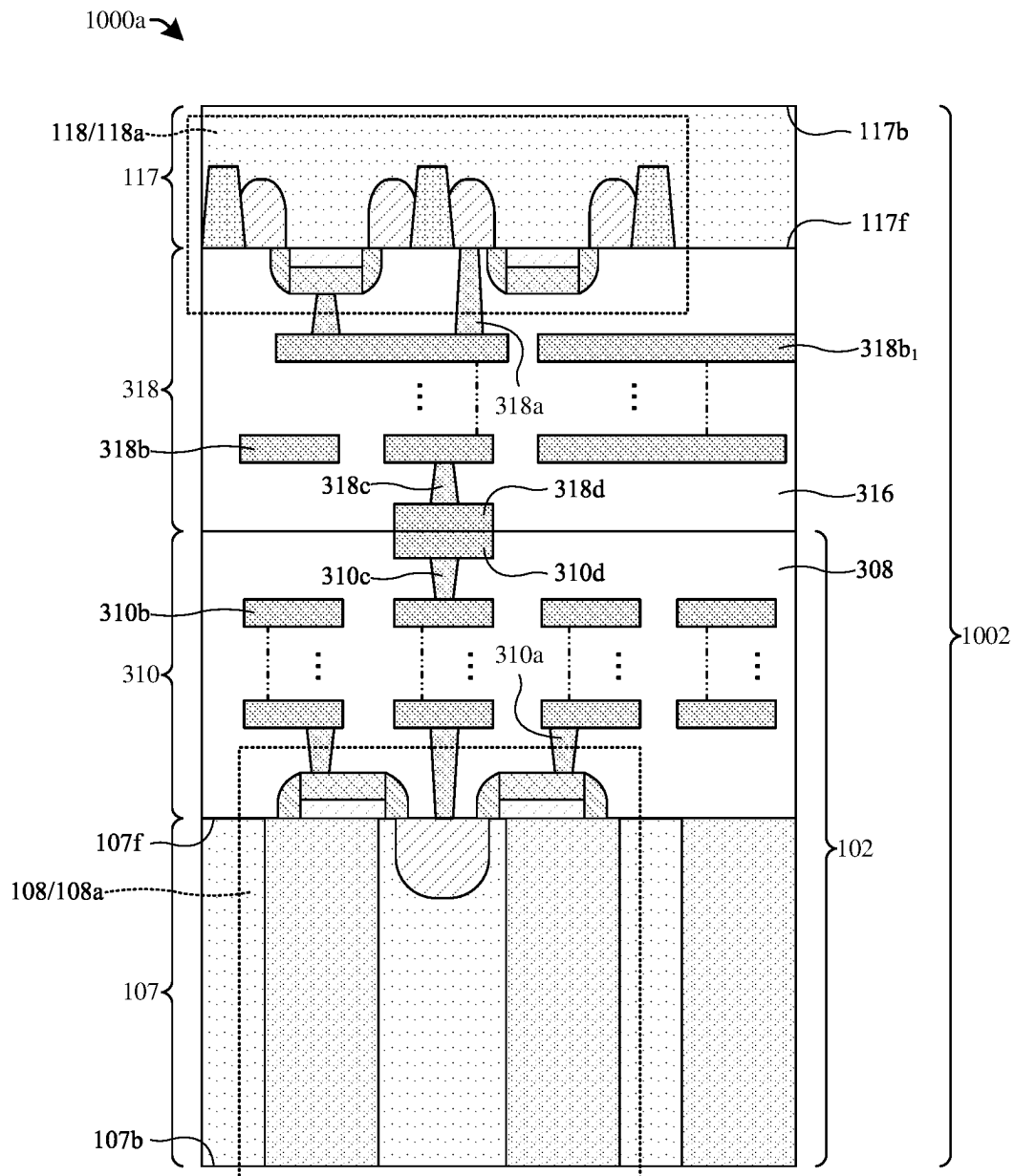


Fig. 10A

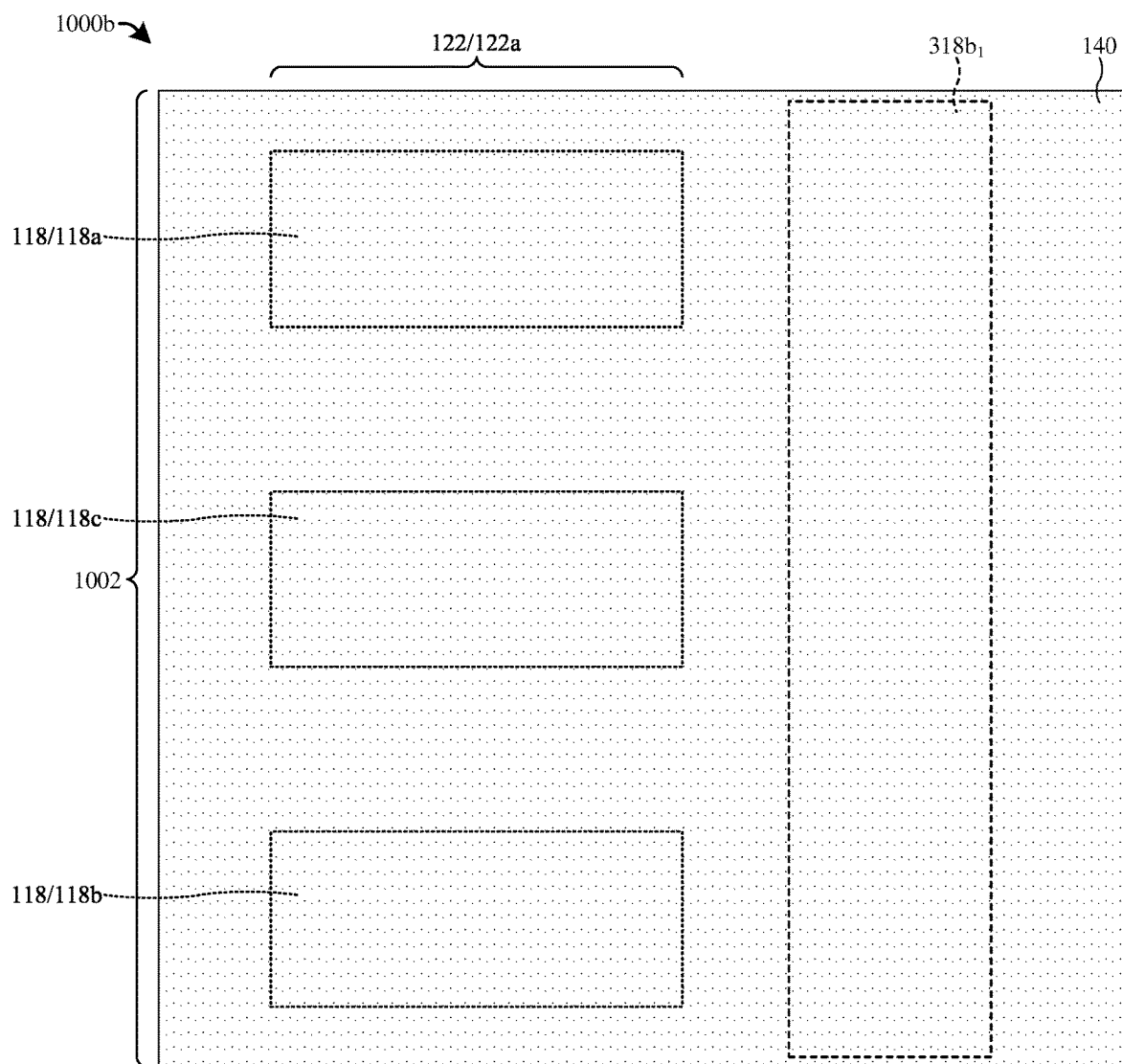


Fig. 10B

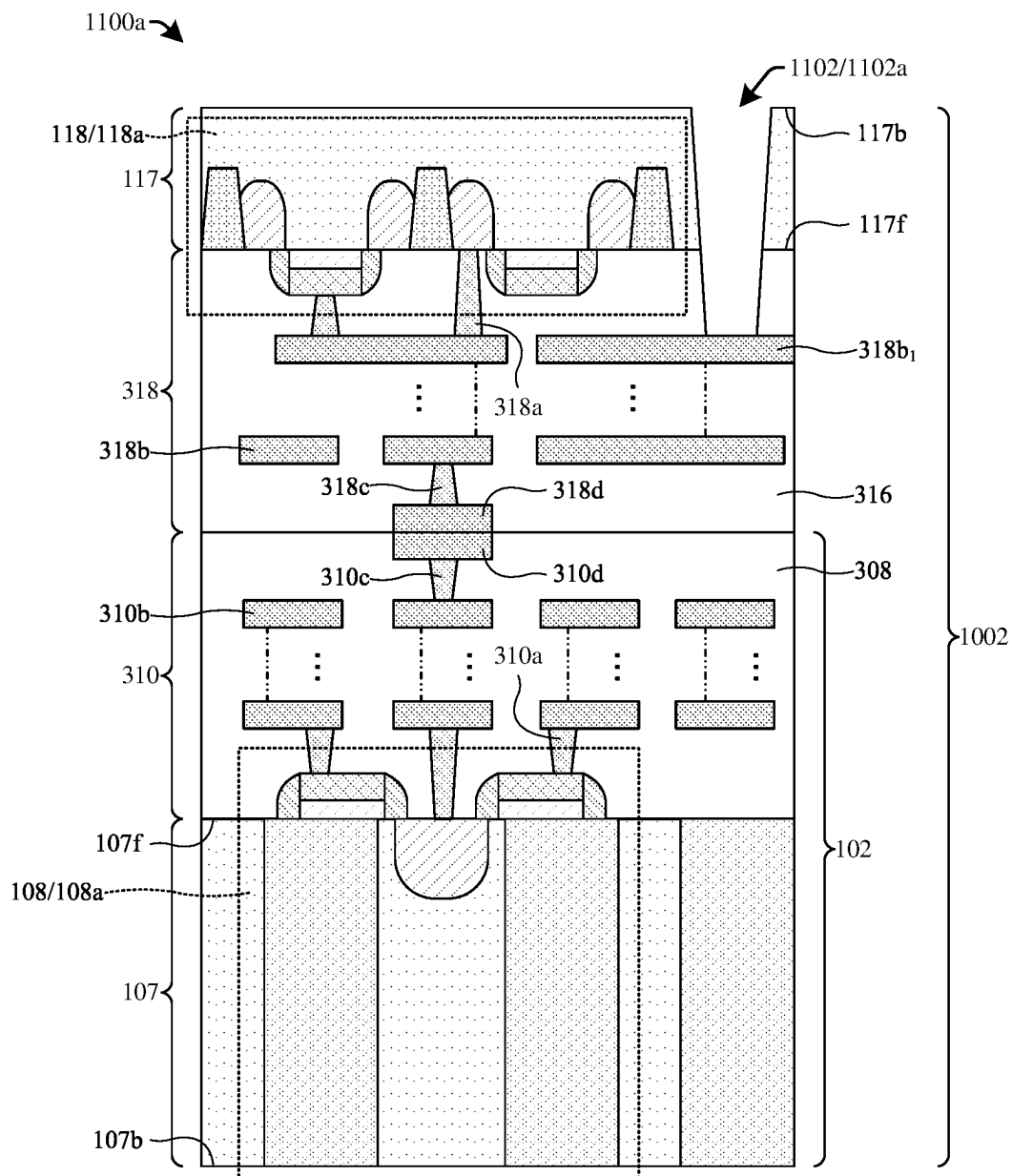


Fig. 11A

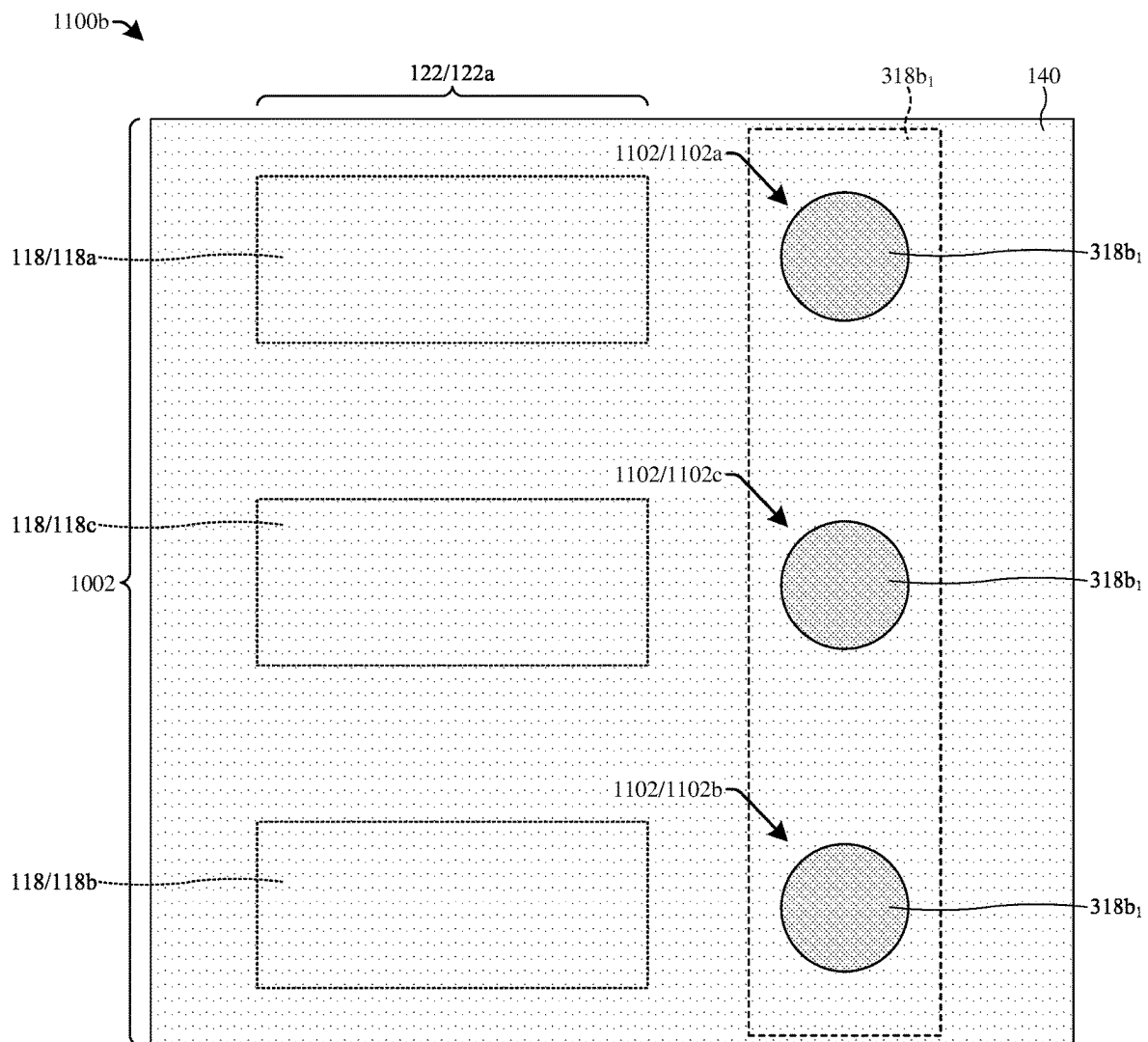


Fig. 11B

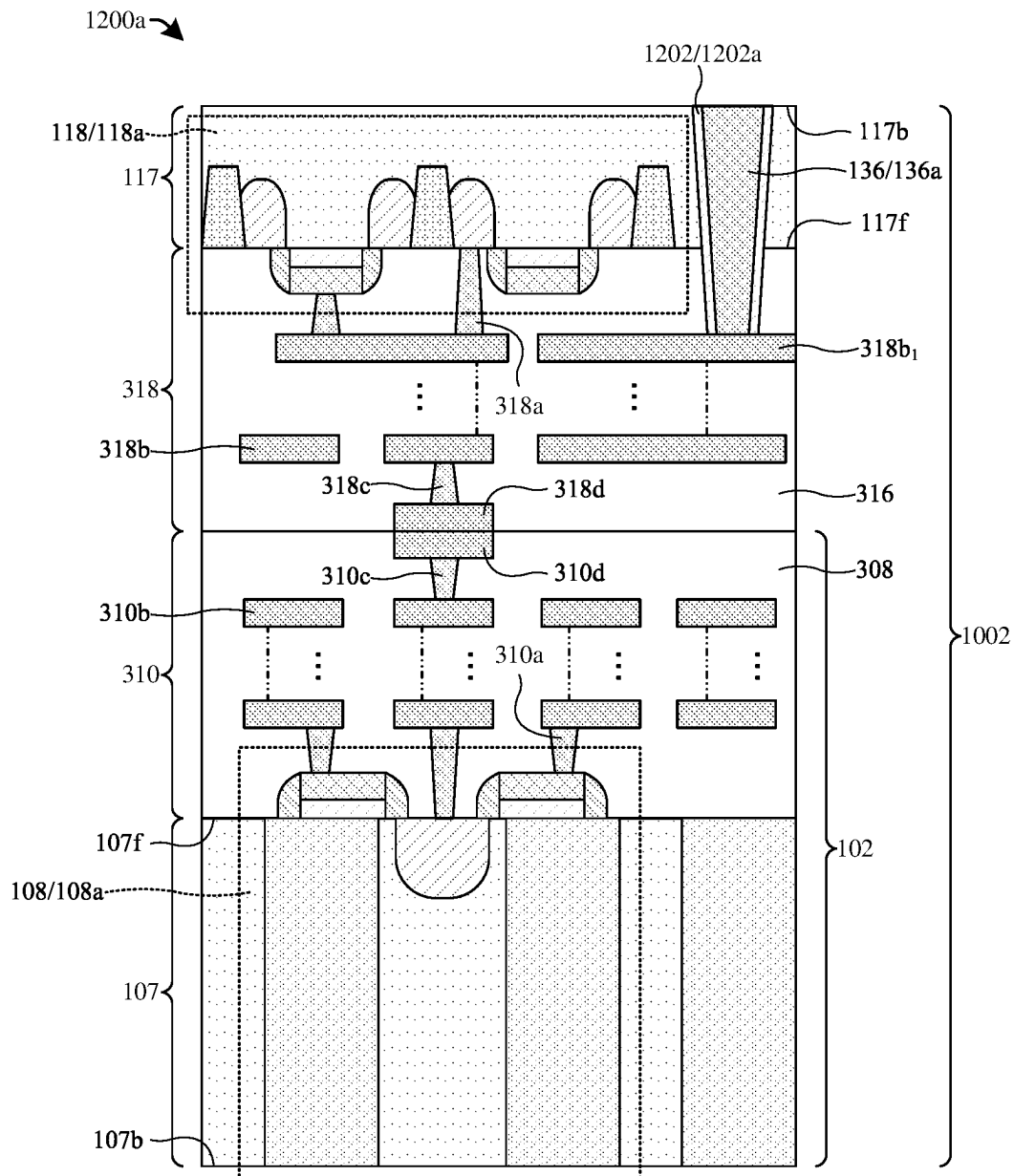


Fig. 12A

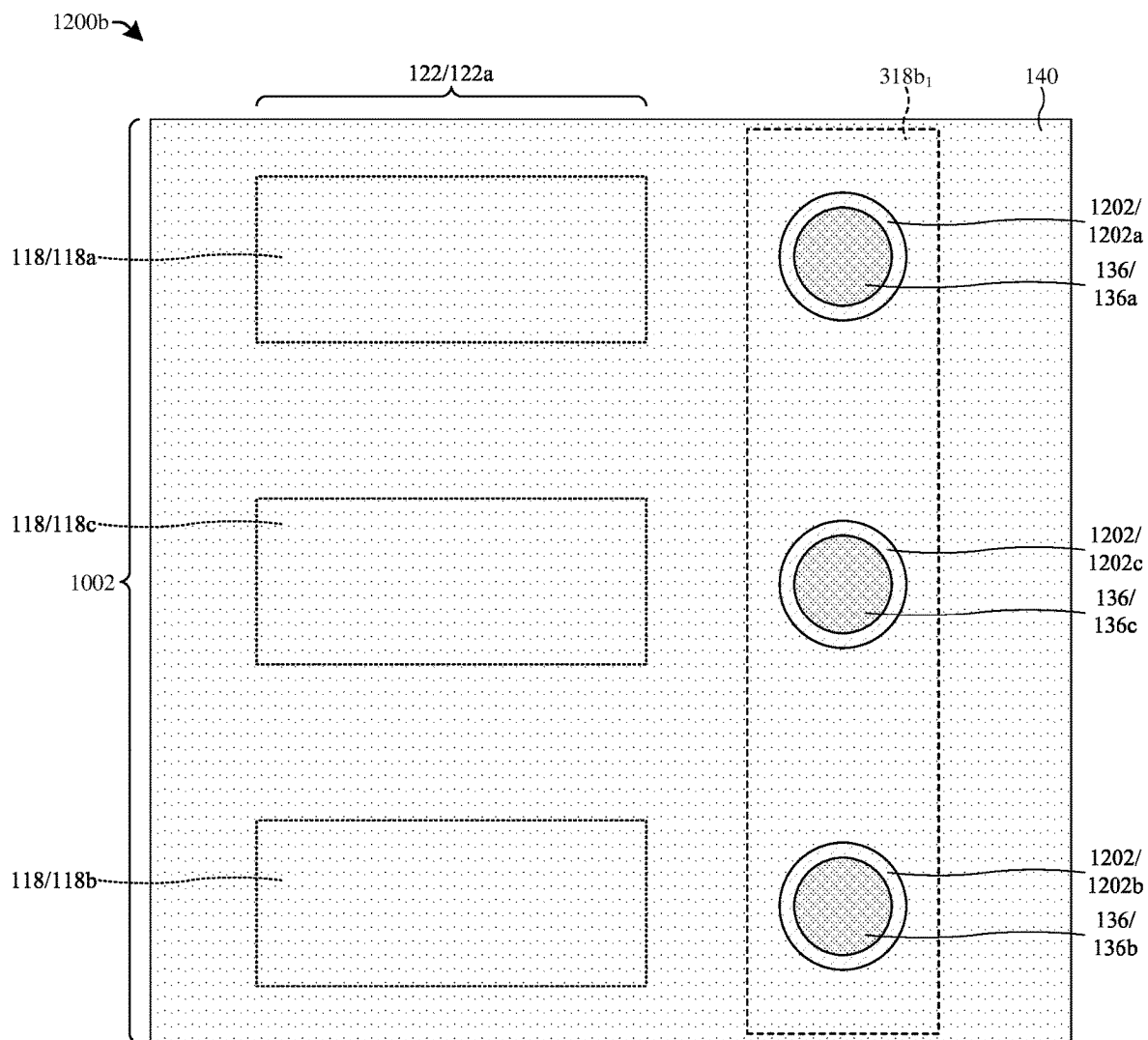


Fig. 12B

Fig. 13A

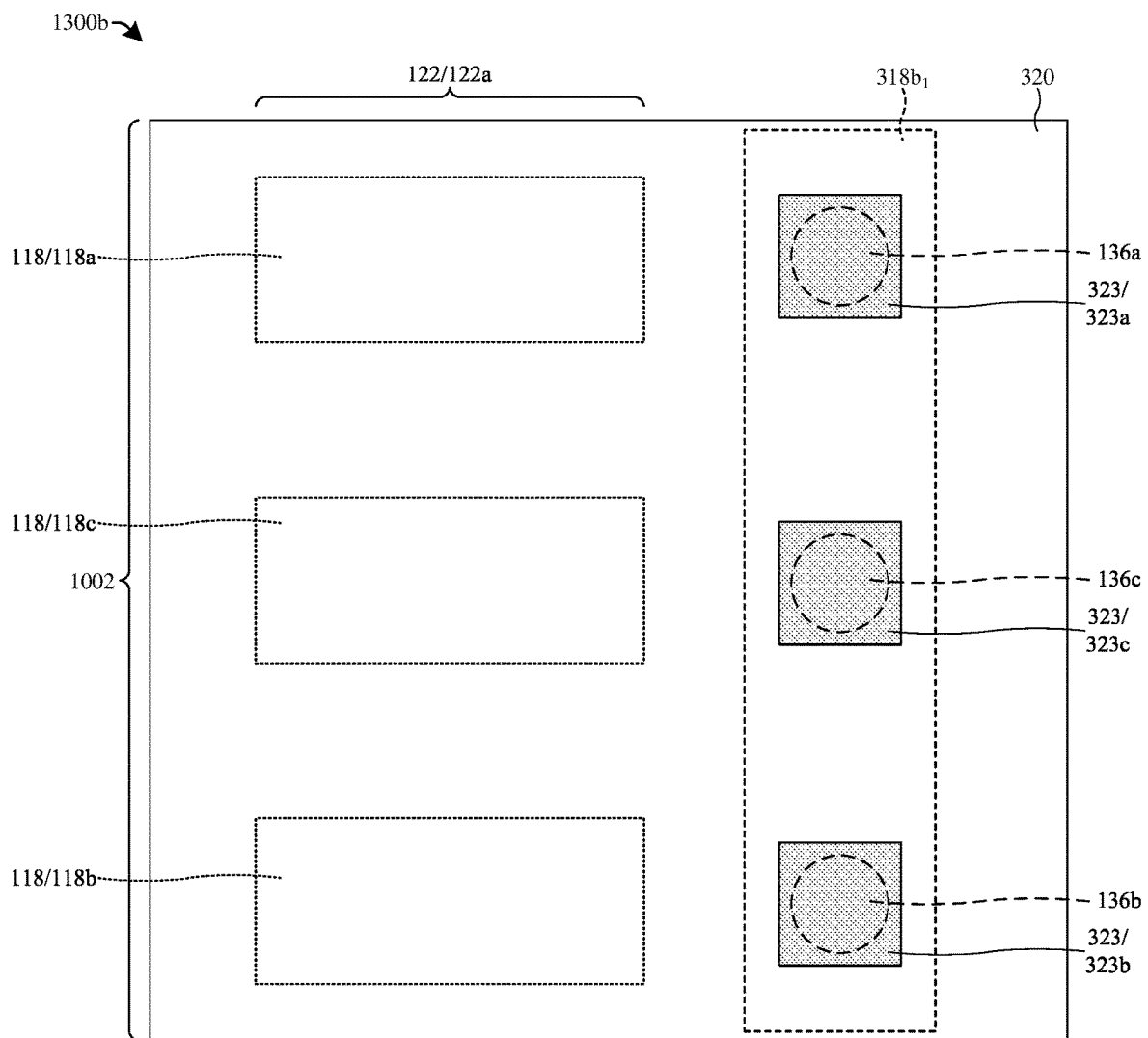
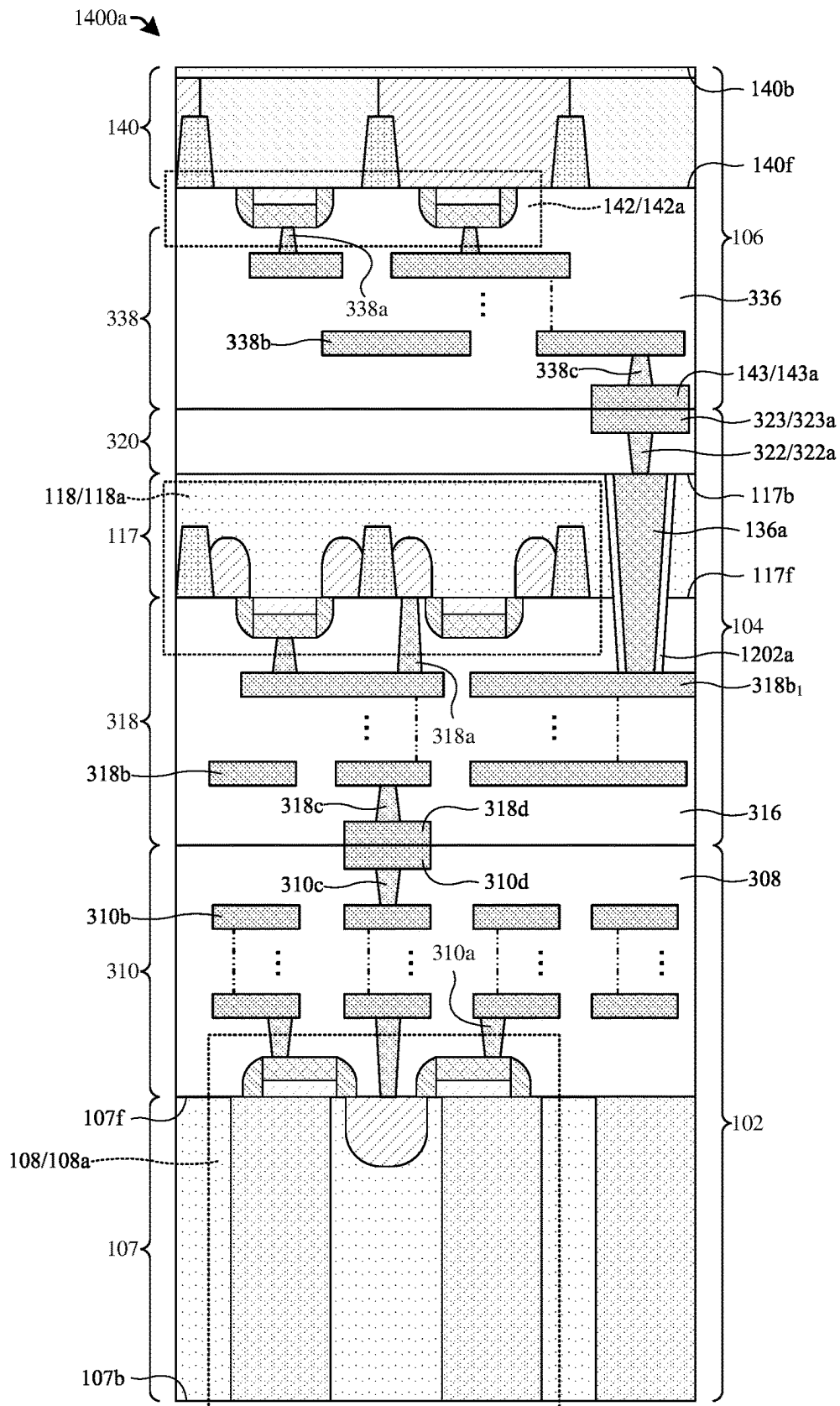


Fig. 13B

**Fig. 14A**

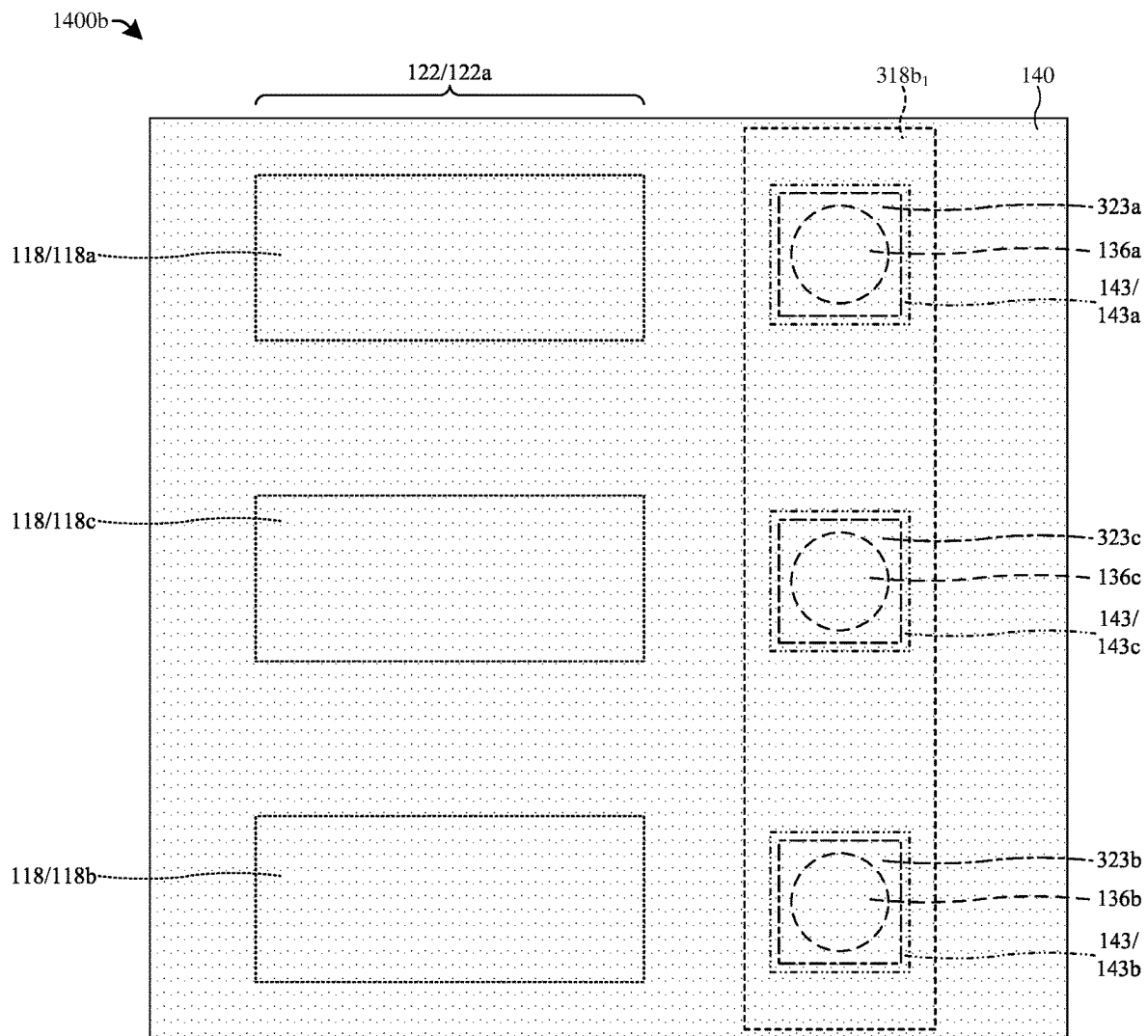
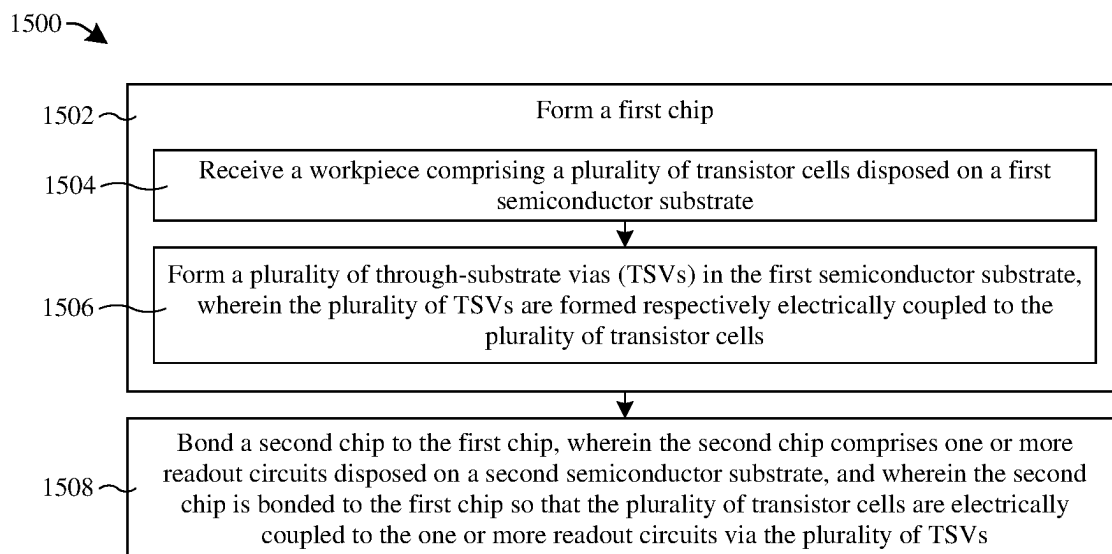


Fig. 14B

**Fig. 15**

HIGH-SPEED READOUT IMAGE SENSOR**REFERENCE TO RELATED APPLICATIONS**

This application claims the benefit of U.S. Provisional Application No. 63/359,981, filed on Jul. 11, 2022 & U.S. Provisional Application No. 63/431,059, filed on Dec. 8, 2022. The contents of the above-referenced Patent applications are hereby incorporated by reference in their entirety.

BACKGROUND

Many modern-day electronic devices (e.g., smartphones, digital cameras, biomedical imaging devices, automotive imaging devices, etc.) comprise image sensors. The image sensors comprise one or more photodetectors (e.g., photodiodes, phototransistors, photoresistors, etc.) configured to absorb incident radiation and output electrical signals corresponding to the incident radiation. Some types of image sensors include charge-coupled device (CCD) image sensors and complementary metal-oxide-semiconductor (CMOS) image sensors. Compared to CCD image sensors, CMOS image sensors are favored due to low power consumption, small size, fast data processing, a direct output of data, and low manufacturing cost. Some types of CMOS image sensors include front-side illuminated (FSI) image sensors and backside illuminated (BSI) image sensors.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A-1B illustrate various views of some embodiments of a high-speed readout image sensor.

FIG. 2 illustrates a layout view of some embodiments of the high-speed readout image sensor.

FIG. 3 illustrates a cross-sectional view of some embodiments of the high-speed readout image sensor.

FIG. 4 illustrates a layout diagram of some embodiments of the high-speed readout image sensor.

FIG. 5 illustrates a layout view of some other embodiments of the high-speed readout image sensor.

FIG. 6 illustrates a schematic diagram of some embodiments of a third chip of some embodiments of the high-speed readout image sensor.

FIG. 7 illustrates a layout diagram of some other embodiments of the high-speed readout image sensor.

FIG. 8 illustrates a schematic diagram of some embodiments of a third chip of some embodiments of the high-speed readout image sensor of FIG. 7.

FIG. 9 illustrates a layout diagram of some other embodiments of the high-speed readout image sensor.

FIGS. 10A-10B through 14A-14B illustrate a series of various views of some embodiments of a method for forming a high-speed readout image sensor.

FIG. 15 illustrates a flowchart of some embodiments of a method for forming a high-speed readout image sensor.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different fea-

tures of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Many portable electronic devices (e.g., cameras, cellular telephones, etc.) include an image sensor for capturing images. One example of such an image sensor is a complementary metal-oxide-semiconductor (CMOS) image sensor (CIS) comprising a first chip, a second chip, and a third chip. The first chip comprises a plurality of photodetector cells. The photodetector cells of the plurality of photodetector cells are disposed in a first array comprising rows and columns (e.g., 1×1 array, 2×2 array, 16×16 array, etc.). Each of the photodetector cells comprises one or more photodetectors disposed in a first semiconductor substrate (e.g., the semiconductor substrate of the first chip) and one or more transfer gates disposed on/over the first semiconductor substrate.

For example, the plurality of photodetector cells may comprise a first photodetector cell and a second photodetector cell. The first photodetector cell may comprise one or more first photodetectors (e.g., 1 photodetector, 2 photodetectors, 16 photodetectors, etc.) and one or more first transfer gates. The second photodetector cell may comprise one or more second photodetectors and one or more second transfer gates.

The second chip comprises a plurality of transistor cells. The transistor cells of the plurality of transistors cells are disposed in a second array comprising rows and columns (e.g., 1×1 array, 2×2 array, 16×16 array, etc.). Each of the plurality of transistor cells comprises a plurality of transistors that are disposed on/over a second semiconductor substrate (e.g., the semiconductor substrate of the second chip) and that are configured to operate one or more corresponding photodetector cells.

For example, the plurality of transistor cells may comprise a first transistor cell and a second transistor cell. The first transistor cell may comprise a first plurality of transistors that are disposed on/over the second semiconductor substrate and that are configured to operate the first photodetector cell. The second transistor cell may comprise a second plurality of transistors that are disposed on/over the second semiconductor substrate and are configured to operate the second photodetector cell.

The third chip comprises a plurality of readout circuits. Each of the plurality of readout circuits comprise one or more readout devices (e.g., transistors) disposed on/over a third semiconductor substrate (e.g., the semiconductor substrate of the third chip). The plurality of readout circuits are electrically coupled to groups of transistor cells of the plurality of transistor cells; and the plurality of readout circuits are configured to receive electrical signals, which correspond to charges accumulated in the plurality of photodetectors, from their corresponding group of transistor cells.

For example, the plurality of readout circuits may comprise a first readout circuit and a second readout circuit. The first readout circuit may comprise one or more first readout devices disposed in the third semiconductor substrate. The first readout circuit is electrically coupled to a first group of transistor cells of the plurality of transistor cells. The first readout circuit is configured to receive electrical signals, which correspond to charges accumulated in the photodetectors that the first group of transistor cells operate, from the first group of transistor cells. In some embodiments, the first group of transistor cells comprises each individual transistor cell that is disposed in a first column of the second array.

The second readout circuit may comprise one or more second readout devices disposed in the third semiconductor substrate. The second readout circuit is electrically coupled to a second group of transistor cells of the plurality of transistor cells. The second readout circuit is configured to receive electrical signals, which correspond to charges accumulated in the photodetectors that the plurality of transistors of the second group of transistor cells operate, from the second group of transistor cells. In some embodiments, the second group of transistor cells comprises each individual transistor cell that is disposed in a second column of the second array.

Further, the second chip comprises an interlayer dielectric (ILD) structure. The plurality of transistor cells are electrically coupled to conductive wires that are disposed in the ILD structure. More specifically, in some embodiments, each of the transistor cells of a given one of the groups of transistor cells is electrically coupled to a corresponding one of the conductive wires. The conductive wires typically run in parallel with one another from a first side of the second chip toward a second side of the second chip.

Moreover, the second chip may comprise a plurality of through-substrate vias (TSVs) that extend vertically through the second semiconductor substrate. The plurality of TSVs extend vertically through the second semiconductor substrate. Typically, each of the conductive wires is electrically coupled to a single one of the plurality of TSVs; and each of the plurality of TSVs provides an electrical connection between one of the conductive wires (of the second chip) and one of the plurality of readout circuits. Further, the plurality of TSVs are generally disposed in a peripheral region of the second chip that extends along an outermost sidewall of the second semiconductor substrate.

For example, a first conductive wire and a second conductive wire may be disposed in the ILD structure. Each of the transistor cells of the first group of transistor cells are electrically coupled to the first conductive wire. Each of the transistor cells of the second group of transistor cells are electrically coupled to the second conductive wire. The plurality of TSVs comprise a first TSV and a second TSV that extend vertically through the second semiconductor substrate. The first TSV is electrically coupled to the first conductive wire; and the first TSV electrically couples the

first conductive wire to the first readout circuit. The second TSV is electrically coupled to the second conductive wire; and the second TSV electrically couples the second conductive wire to the second readout circuit. The first TSV and the second TSV are laterally spaced from one another. The first TSV and the second TSV are both disposed in the peripheral region of the second chip.

Because the plurality of transistor cells are electrically coupled to the conductive wires, and because the plurality of TSVs are disposed in the peripheral region of the second chip, the conductive wires are relatively long (e.g., extend a relatively long lateral distance across the second chip). More specifically, the plurality of transistor cells generally comprise a large number of transistor cells, thereby resulting in the plurality of transistor cells consuming a large area of the second chip. Further, because each of the plurality of TSVs provides an electrical connection between one of the conductive wires (of the second chip) and one of the plurality of readout circuits, the conductive wires are relatively long to ensure the plurality of transistor cells may be electrically coupled to the plurality of TSVs. Because the conductive wires are relatively long, and because each of the plurality of TSVs provides an electrical connection between one of the conductive wires (of the second chip) and one of the plurality of readout circuits, a typical image sensor may have a relatively high RC delay, thereby resulting in the typical image sensor having a slow readout speed (e.g., long output signal settling time, decreased chip speed, etc.).

As another more specific example, in some embodiments, the plurality of transistor cells may comprise the first transistor cell, the second transistor cell, a third transistor cell, and a fourth transistor cell. The first transistor cell and the second transistor cell may be disposed in the first column of the second array; and the third transistor cell and the fourth transistor cell may be disposed in the second column of the second array. The first transistor cell and the third transistor cell may be disposed in a first row of the second array; and the second transistor cell and the fourth transistor cell may be disposed in a second row of the second array. The first row of the second array may be an outermost row on a first side of the second array; and the second row of the second array may be an outermost row on a second side of the second array opposite the first side of the second array (e.g., the first row of the second array is spaced further from the second row of the second array than any other of the rows of the second array). The first TSV and the second TSV may be disposed outside the second array and on a same side of the second array (e.g., in the peripheral region which is outside the second array and along the first side of the second array). The first conductive wire may be electrically coupled to the first transistor cell, the second transistor cell, and the first TSV; and the second conductive wire may be electrically coupled to the third transistor cell, the fourth transistor cell, and the second TSV. The first TSV may electrically couple the first conductive wire to the first readout circuit; and the second TSV may electrically couple the second conductive wire to the first readout circuit.

Because the first transistor cell, the second transistor cell, and the first TSV are electrically coupled to the first conductive wire, and because of the relatively large spacing between such structures (e.g., the first transistor cell is spaced relatively far from the second transistor cell, and the first TSV is disposed in the peripheral region which is along the first side of the second array), the first conductive wire is relatively long. For a similar reason, the second conductive wire is also relatively long. Because the first and second conductive wires are relatively long, and because the first

TSV provides an electrical connection between the first conductive wire and (only) the first readout circuit and the second TSV provides an electrical connection between the second conductive wire and (only) the second readout circuit, the typical image sensor may have a relatively high RC delay, thereby resulting in the typical image sensor having the slow readout speed.

Various embodiments of the present disclosure are related to a high-speed readout image sensor. The high-speed readout image sensor comprises a first chip and a second chip that is bonded to the first chip. The first chip comprises a first semiconductor substrate. The first chip also comprises a first transistor cell and a second transistor cell disposed in a transistor cell array. The first transistor cell comprises a first plurality of transistors disposed on the first semiconductor substrate. The first transistor cell is configured to operate a first photodetector cell. The second transistor cell comprises a second plurality of transistors disposed on the first semiconductor substrate. The second transistor cell is configured to operate a second photodetector cell. In some embodiments, a third chip of the high-speed readout image sensor comprises the first photodetector cell and the second photodetector cell.

The first chip also comprises a first through-substrate via (TSV) that extends vertically through the first semiconductor substrate. The first transistor cell is electrically coupled to the first TSV. The first transistor cell is configured to provide a first signal to the first TSV that corresponds to a number of charges accumulated in a photodetector of the first photodetector cell.

The first chip also comprises a second TSV that extends vertically through the first semiconductor substrate. The second transistor cell is electrically coupled to the second TSV. The second transistor cell is configured to provide a second signal to the second TSV that corresponds to a number of charges accumulated in a photodetector of the second photodetector cell.

The second chip comprises a second semiconductor substrate. The second chip also comprises an interlayer dielectric (ILD) structure disposed vertically between the second semiconductor substrate and first semiconductor substrate. The second chip also comprises a readout circuit that is electrically coupled to the first TSV and the second TSV. The readout circuit is configured to receive the first signal and the second signal. The readout circuit is disposed laterally, at least partially, between the first TSV and the second TSV.

Because the readout circuit is disposed laterally, at least partially, between the first TSV and the second TSV, overall lengths of the conductive paths that electrically couple the first transistor cell and the second transistor cell to the readout circuit may be relatively small (e.g., a total length of the conductive paths may be less than (e.g., 50% less than) a total length of corresponding conductive paths of the typical image sensor). Because the overall lengths of the conductive paths that electrically couple the first transistor cell and the second transistor cell to the readout circuit are relatively small, the high-speed readout image sensor may have a relatively low RC delay, thereby resulting in the high-speed readout image sensor having a fast readout speed (e.g., quick output signal settling time, increased chip speed, etc.).

FIGS. 1A-1B illustrate various views 100a-100b of some embodiments of a high-speed readout image sensor. FIG. 1A illustrates a schematic diagram 100a of some embodiments of the high-speed readout image sensor. FIG. 1B illustrates a layout diagram 100b of some embodiments of the high-speed readout image sensor.

As shown in the various views 100a-100b of FIGS. 1A-1B, the high-speed readout image sensor comprises a first chip 102, a second chip 104, and a third chip 106. The first chip 102 is bonded to the second chip 104, and the second chip 104 is bonded to the third chip 106. In some embodiments, the second chip 104 is disposed vertically between (e.g., sandwiched between) the first chip 102 and the third chip 106.

The first chip 102 comprises a first semiconductor substrate 107. The first semiconductor substrate 107 may comprise any type of semiconductor body (e.g., monocrystalline silicon/CMOS bulk, germanium (Ge), a group III-V semiconductor material, silicon-germanium (SiGe), silicon on insulator (SOI), etc.). The first chip 102 comprises a plurality of photodetector cells 108 disposed on the first semiconductor substrate 107. For clarity in the figures, only some of the plurality of photodetector cells 108 are labeled. The plurality of photodetector cells 108 are laterally spaced. In some embodiments, the photodetector cells of the plurality of photodetector cells 108 are disposed in a first array. The first array comprises a first plurality of rows 110 and a first plurality of columns 112. Each of the photodetector cells of the plurality of photodetector cells 108 comprises one or more photodetectors 114 and one or more transfer gates 116.

The second chip 104 comprises a second semiconductor substrate 117. The second semiconductor substrate 117 may comprise any type of semiconductor body (e.g., monocrystalline silicon/CMOS bulk, germanium (Ge), a group III-V semiconductor material, silicon-germanium (SiGe), silicon on insulator (SOI), etc.). The second chip 104 comprises a plurality of transistor cells 118 disposed on the second semiconductor substrate 117. The plurality of transistor cells 118 are laterally spaced. In some embodiments, the transistor cells of the plurality of transistor cells 118 are disposed in a second array. The second array comprises a second plurality of rows 120 and a second plurality of columns 122. Each of the transistor cells of the plurality of transistor cells 118 comprises a plurality of transistors 124.

The plurality of transistor cells 118 are electrically coupled to the plurality of photodetector cells 108, respectively. For example, the plurality of photodetector cells 108 may comprise a first photodetector cell 108a, a second photodetector cell 108b, and so forth. The plurality of transistor cells 118 may comprise a first transistor cell 118a, a second transistor cell 118b, and so forth. The first transistor cell 118a is electrically coupled to the first photodetector cell 108a; the second transistor cell 118b is electrically coupled to the second photodetector cell 108b; and so forth.

In some embodiments, the plurality of transistor cells 118 are electrically coupled to floating diffusion nodes of the plurality of photodetector cells 108, respectively. In further embodiments, the one or more photodetectors 114 are configured to absorb incident radiation (e.g., light) and generate electrical signals corresponding to the incident radiation. In yet further embodiments, the one or more transfer gates 116 are configured to transfer accumulated charges from a corresponding photodetector to a corresponding floating diffusion node. For example, a first one of the one or more transfer gates 116 of the first photodetector cell 108a is configured to transfer charges accumulated in a first one of the one or more photodetectors 114 of the first photodetector cell 108a to a floating diffusion node 126 of the first photodetector cell 108a.

The plurality of transistor cells 118 are configured to operate (e.g., access, select, reset, etc.) the plurality of photodetector cells 108, respectively. For example, the first transistor cell 118a is configured to operate the first photo-

detector cell **108a**; the second transistor cell **118b** is configured to operate the second photodetector cell **108b**; and so forth. The plurality of transistor cells **118** correspond to the plurality of photodetector cells **108** in which they are configured to operate. For example, the first transistor cell **118a** corresponds to the first photodetector cell **108a**; the second transistor cell **118b** corresponds to the second photodetector cell **108b**; and so forth.

In some embodiments, the plurality of transistors comprise a first transistor **124a**, a second transistor **124b**, and a third transistor **124c**. In some embodiments, the first transistor **124a** is a reset transistor. In some embodiments, the second transistor **124b** is a source-follower transistor. In some embodiments, the third transistor **124c** is a select transistor (e.g., row select transistor).

In some embodiments, a first source/drain node **128** of the first transistor **124a** is electrically coupled to a gate **130** of the second transistor **124b**. In further embodiments, the first source/drain node **128** and the gate **130** are electrically coupled to the floating diffusion node **126** of their corresponding photodetector cell. For example, the first source/drain node **128** of the first transistor **124a** of the first transistor cell **118a** and the gate **130** of the second transistor **124b** of the first transistor cell **118a** are both electrically coupled to the floating diffusion node **126** of the first photodetector cell **108a**. In some embodiments, a first source/drain node **132** of the second transistor **124b** is electrically coupled to a first source/drain node **134** of the third transistor **124c**.

The second chip **104** comprises a plurality of through-substrate vias (TSVs) **136**. For clarity in the drawings, only some of the plurality of TSVs **136** are labeled. The plurality of TSVs **136** extend vertically through the second semiconductor substrate **117**. The plurality of TSVs **136** are electrically coupled to the plurality of transistor cells **118**, respectively. For example, the plurality of TSVs **136** comprise a first TSV **136a**, a second TSV **136b**, and so forth. The first TSV **136a** is electrically coupled to the first transistor cell **118a**; the second TSV **136b** is electrically coupled to the second transistor cell **118b**; and so forth. In some embodiments, the plurality of TSVs **136** are electrically coupled to a second source/drain node **138** of the third transistor **124c** of their corresponding transistor cell. For example, the first TSV **136a** may be electrically coupled to the second source/drain node **138** of the third transistor **124c** of the first transistor cell **118a**; the second TSV **136b** may be electrically coupled to a second source/drain node **138** of the third transistor **124c** of the second transistor cell **118b**; and so forth. In some embodiments, the plurality of TSVs **136** are or comprise, for example, copper (Cu), aluminum (Al), gold (Au), silver (Ag), platinum (Pt), polysilicon, some other conductive material, or a combination of the foregoing.

The plurality of transistor cells **118** are configured to provide signals to the plurality of TSVs **136** that correspond to a number of charges accumulated in the one or more photodetectors of their corresponding photodetector cell. For example, the first transistor cell **118a** is configured to provide one or more first signals to the first TSV **136a** that correspond to charges accumulated in the one or more photodetectors **114** of the first photodetector cell **108a**; the second transistor cell **118b** is configured to provide one or more second signals to the second TSV **136b** that correspond to charges accumulated in the one or more photodetectors **114** of the second photodetector cell **108b**; and so forth.

In some embodiments, the first TSV **136a** and the second TSV **136b** are spaced further apart than any other TSV of the plurality of TSVs **136** that are disposed in a same column as

the first TSV **136a** and the second TSV **136b**. In other embodiments, the first TSV **136a** may neighbor the second TSV **136b**. In further embodiments, the plurality of TSVs **136** may be laterally spaced from each of the edges (e.g., outermost sidewalls) of the second semiconductor substrate **117** by at least about 500 micrometers (μm). In yet further embodiments, the plurality of TSVs **136** may be laterally spaced from each of the edges (e.g., outermost sidewalls) of the second semiconductor substrate **117** by at least about 2000 μm .

The third chip **106** comprises a third semiconductor substrate **140**. The third semiconductor substrate **140** may comprise any type of semiconductor body (e.g., monocrystalline silicon/CMOS bulk, germanium (Ge), a group III-V semiconductor material, silicon-germanium (SiGe), silicon on insulator (SOI), etc.). The third chip **106** comprises a first plurality of readout circuits **142**. In some embodiments, each of the first plurality of readout circuits **142** may comprise an amplifier circuit **144**. In further embodiments, each of the first plurality of readout circuits **142** may comprise an analog-to-digital converter (ADC) **146**. In some embodiments, the amplifier circuit **144** is electrically coupled to the ADC **146**.

The third chip **106** also comprises a first plurality of conductive bond structures **143**. The first plurality of conductive bond structures **143** are electrically coupled to the plurality of TSVs **136**, respectively. For example, a first conductive bond structure **143a** is electrically coupled to the first TSV **136a**; a second conductive bond structure **143b** is electrically coupled to the second TSV **136b**; and so forth. For clarity in the figures, only some of the first plurality of conductive bond structures **143** are labeled. In some embodiments, the first plurality of conductive bond structures **143** may be or comprise, for example, copper (Cu), aluminum (Al), tungsten (W), gold (Au), some other conductive material, or a combination of the foregoing.

In some embodiments, the first plurality of conductive bond structures **143** are disposed in a third array. The third array comprises a third plurality of rows **150** and a third plurality of columns **148**. For clarity in the figures, the plurality of TSVs **136** are illustrated in phantom (via dotted lines) on the third chip **106**. It will be appreciated that the plurality of TSVs **136** may also be disposed in an array (e.g., a fourth array comprising a fourth plurality of rows and a fourth plurality of columns), as shown in the layout diagram **100b** of FIG. 1B.

The first plurality of conductive bond structures **143** are electrically coupled to the first plurality of readout circuits **142**. More specifically, in some embodiments, the conductive bond structures of the first plurality of conductive bond structures **143** of a given column of the third plurality of columns **148** are electrically coupled to a corresponding one of the first plurality of readout circuits **142**. For example, in some embodiments, each of the first plurality of conductive bond structures **143** that are disposed in a first column **148a** of the third plurality of columns **148** are electrically coupled to a first readout circuit **142a** of the first plurality of readout circuits **142**; each of the first plurality of conductive bond structures **143** that are disposed in a second column **148b** of the third plurality of columns **148** are electrically coupled to a second readout circuit **142b** of the first plurality of readout circuits **142**; and so forth. It will be appreciated that the ellipsis symbols (. . .) shown in figures illustrate that the first chip **102**, the second chip **104**, and/or the third chip **106** may be larger than illustrated and may comprise additional

iterations of the features (e.g., additional TSVs, additional photodetectors cells, additional transistors cells, etc.) illustrated in the figures.

In some embodiments, the conductive bond structures of the first plurality of conductive bond structures **143** of a given column of the third plurality of columns **148** are electrically coupled via a corresponding conductive feature (e.g., conductive wire) of the third chip **106**. For example, in some embodiments, each of the first plurality of conductive bond structures **143** that are disposed in the first column **148a** are electrically coupled via a first conductive feature of the third chip **106** (e.g., first conductive wire of an interconnect structure of the third chip); each of the first plurality of conductive bond structures **143** that are disposed in the second column **148b** are electrically coupled via a second conductive feature of the third chip **106** (e.g., second conductive wire of the interconnect structure of the third chip); and so forth. It will be appreciated that the solid lines extending between some of the first plurality of conductive bond structures **143** illustrates that such conductive bond structures are electrically coupled. It will also be appreciated that the dotted lines extending from some of the first plurality of conductive bond structures **143** to the first plurality of readout circuits **142** illustrate that such conductive bond structures are electrically coupled to the first plurality of readout circuits **142**. It will also be appreciated that the dotted lines extending from some of the first plurality of conductive bond structures **143** to the first plurality of readout circuits **142** illustrate that such conductive bond structures may be electrically coupled to additional conductive bond structures (not shown, but appreciated that they may exist due to the ellipsis symbols) that are disposed laterally between such conductive bond structures and the first plurality of readout circuits **142**.

Because the first plurality of conductive bond structures **143** are electrically coupled to the plurality of TSVs **136** and electrically coupled to the first plurality of readout circuits **142**, the plurality of transistor cells **118** are also configured to provide the signals (that correspond to a number of charges accumulated in the one or more photodetectors of their corresponding photodetector cell) to the first plurality of readout circuits **142**. More specifically, in some embodiments, the transistor cells of the plurality of transistor cells **118** of a given column of the second plurality of columns **122** are configured to provide the signals to a corresponding readout circuit. For example, the first transistor cell **118a** is configured to provide the one or more first signals that correspond to charges accumulated in the one or more photodetectors **114** of the first photodetector cell **108a** to the first readout circuit **142a** (via the first TSV **136a**); the second transistor cell **118b** is configured to provide the one or more second signals that correspond to charges accumulated in the one or more photodetectors **114** of the second photodetector cell **108b** to the first readout circuit **142a** (via the second TSV **136b**); and so forth.

Also shown in the layout diagram **100b** of FIG. 1B, the first plurality of readout circuits **142** are laterally disposed, at least partially, between the first TSV **136a** and the second TSV **136b**. Because the first plurality of readout circuits **142** are disposed laterally, at least partially, between the first TSV **136a** and the second TSV **136b**, overall lengths of the conductive paths that electrically couple the plurality of transistor cells **118** to their corresponding one of the first plurality of readout circuits **142** may be relatively small (e.g., a total length of the conductive paths may be less than (e.g., 50% less than) a total length of corresponding conductive paths of a typical image sensor). Because the overall

lengths of the conductive paths are relatively small, and because the first plurality of readout circuits **142** are configured to receive signals (that correspond to charges accumulated in photodetectors) from the plurality of transistor cells **118**, the high-speed readout image sensor may have a relatively low RC delay, thereby resulting in the high-speed readout image sensor having a fast readout speed (e.g., quick output signal settling time, increased chip speed, etc.).

For example, because of the first TSV **136a**, and because the first readout circuit **142a** is disposed laterally between the first TSV **136a** and the second TSV **136b**, a relatively small first conductive path exists between the first transistor cell **118a** and the first readout circuit **142a**. Further, because of the second TSV **136b**, and because the first readout circuit **142a** is disposed laterally between the first TSV **136a** and the second TSV **136b**, a relatively small second conductive path exists between the second transistor cell **118b** and the first readout circuit **142a**. Therefore, in comparison to a typical image sensor (e.g., one that only comprises one TSV per column of transistor cells), the first readout circuit **142a** may receive the one or more first signals (provided via the first transistor cell **118a**) and the one or more second signals (provided via the second transistor cell **118b**) more quickly, thereby resulting in a lower RC delay.

FIG. 2 illustrates a layout view **200** of some embodiments of the high-speed readout image sensor. While the layout view **200** illustrates the first chip **102**, the second chip **104**, and the third chip **106** laterally spaced from one another, it will be appreciated that this is for clarity in the figures and that the first chip **102**, the second chip **104**, and the third chip **106** may be vertically stacked and bonded together.

As shown in the layout view **200** of FIG. 2, the first chip **102** comprises the first photodetector cell **108a**. The first photodetector cell **108a** comprises one or more photodetectors **202**. In some embodiments, the one or more photodetectors **202** comprise four individual photodetectors, as shown in the layout view **200** of FIG. 2. The one or more photodetectors **202** are configured to absorb the incident radiation (e.g., light) and generate electrical signals corresponding to the incident radiation. It will be appreciated that the one or more photodetectors **202** may comprise any number of individual photodetectors (e.g., 1 photodetector, 2 photodetectors, 16 photodetectors, etc.). It will also be appreciated that, in some embodiments, the one or more photodetectors **202** may be substantially the same (e.g., comprise similar structural features (and/or have a substantially similar layout)) as the one or more photodetectors **114**, or vice versa.

The first photodetector cell **108a** comprises one or more transfer gates **204** and a floating diffusion node **206**. In some embodiments, the one or more transfer gates **204** comprise a same number of individual transfer gates as the one or more photodetectors **202** comprises individual photodetectors. For example, as shown in the layout view **200** of FIG. 2, the one or more transfer gates **204** may comprise four individual transfer gates. It will be appreciated that the one or more transfer gates **204** may comprise any number of individual transfer gates (e.g., 1 transfer gate, 2 transfer gates, 16 transfer gates, etc.). It will also be appreciated that, in some embodiments, the one or more transfer gates **204** may be substantially the same (e.g., comprise similar structural features (and/or have a substantially similar layout)) as the one or more transfer gates **116**, or vice versa.

The one or more transfer gates **204** are configured to transfer accumulated charges from a corresponding one of the one or more photodetectors **202** to the floating diffusion node **206**. For example, a first individual transfer gate of the

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one or more transfer gates **204** is configured to transfer accumulated charges from a first individual photodetector of the one or more photodetectors **202** to the floating diffusion node **206**; a second individual transfer gate of the one or more transfer gates **204** is configured to transfer accumulated charges from a second individual photodetector of the one or more photodetectors **202** to the floating diffusion node **206**; and so forth.

The first chip **102** may also comprise a third photodetector cell **108c** of the plurality of photodetector cells **108**. In some embodiments, the first photodetector cell **108a** neighbors the third photodetector cell **108c**. In other embodiments, one or more other photodetector cells are disposed laterally between the first photodetector cell **108a** and the third photodetector cell **108c**. In further embodiments, the first photodetector cell **108a** and the third photodetector cell **108c** are disposed in a first column **112a** of the first plurality of columns **112**. In yet further embodiments, the first photodetector cell **108a** may be spaced further from the third photodetector cell **108c** than any other photodetector cells that are disposed in the first column **112a**.

The third photodetector cell **108c** comprises one or more photodetectors **208**. In some embodiments, the one or more photodetectors **208** comprise four individual photodetectors, as shown in the layout view **200** of FIG. 2. The one or more photodetectors **208** are configured to absorb the incident radiation (e.g., light) and generate electrical signals corresponding to the incident radiation. It will be appreciated that the one or more photodetectors **208** may comprise any number of individual photodetectors (e.g., 1 photodetector, 2 photodetectors, 16 photodetectors, etc.). It will also be appreciated that, in some embodiments, the one or more photodetectors **208** may be substantially the same (e.g., comprise similar structural features (and/or have a substantially similar layout)) as the one or more photodetectors **114**, or vice versa.

The third photodetector cell **108c** comprises one or more transfer gates **210** and a floating diffusion node **212**. In some embodiments, the one or more transfer gates **210** comprise a same number of individual transfer gates as the one or more photodetectors **208** comprises individual photodetectors. For example, as shown in the layout view **200** of FIG. 2, the one or more transfer gates **210** may comprise four individual transfer gates. It will be appreciated that the one or more transfer gates **210** may comprise any number of individual transfer gates (e.g., 1 transfer gate, 2 transfer gates, 16 transfer gates, etc.). It will also be appreciated that, in some embodiments, the one or more transfer gates **210** may be substantially the same (e.g., comprise similar structural features (and/or have a substantially similar layout)) as the one or more transfer gates **116**, or vice versa.

The one or more transfer gates **210** are configured to transfer accumulated charges from a corresponding one of the one or more photodetectors **208** to the floating diffusion node **212**. For example, a first individual transfer gate of the one or more transfer gates **210** is configured to transfer accumulated charges from a first individual photodetector of the one or more photodetectors **208** to the floating diffusion node **212**; a second individual transfer gate of the one or more transfer gates **210** is configured to transfer accumulated charges from a second individual photodetector of the one or more photodetectors **208** to the floating diffusion node **212**; and so forth.

In some embodiments, the third photodetector cell **108c** comprises substantially the same features (e.g., structural features, such as photodetectors, transfer gates, floating diffusion nodes, etc.) and/or has substantially the same

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layout as the first photodetector cell **108a**, as shown in the layout view **200** of FIG. 2. In further embodiments, each of the photodetector cells of the plurality of photodetector cells **108** comprise substantially the same features and/or have substantially the same layout.

The second chip **104** comprises the first transistor cell **118a**. In some embodiments, the first transistor cell **118a** comprises a first plurality of transistors **214**. The first plurality of transistors **214** comprises a first transistor **214a**, a second transistor **214b**, and a third transistor **214c**. In some embodiments, the first transistor **214a** is a reset transistor. In some embodiments, the second transistor **214b** is a source-follower transistor. In some embodiments, the third transistor **214c** is a select transistor (e.g., row select transistor).

The first transistor **214a**, the second transistor **214b**, and the third transistor **214c** each comprise a gate electrode structure **216** overlying a gate dielectric structure (not shown). The gate dielectric structure overlies the second semiconductor substrate **117**. In some embodiments, the gate electrode structure **216** is or comprises, for example, polysilicon, a metal (e.g., aluminum (Al), copper (Cu), titanium (Ti), tantalum (Ta), tungsten (W), molybdenum (Mo), cobalt (Co), or the like), some other conductive material, or a combination of the foregoing. In further embodiments, the gate dielectric structure is or comprises, for example, an oxide (e.g., silicon dioxide (SiO₂)), a high-k dielectric material (e.g., hafnium oxide (HfO), tantalum oxide (TaO), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), aluminum oxide (AlO), zirconium oxide (ZrO), some other dielectric material with a dielectric constant greater than about 3.9), some other dielectric material, or a combination of the foregoing.

The first transistor **214a**, the second transistor **214b**, and the third transistor **214c** each comprise source/drain regions **218** disposed on opposite sides of a corresponding gate electrode structure. For example, the first transistor **214a** comprises source/drain regions **218** disposed on opposite sides of the gate electrode structure of the first transistor **214a**; the second transistor **214b** comprises source/drain regions **218** disposed on opposite sides of the gate electrode structure of the second transistor **214b**; and so forth. The source/drain regions **218** are disposed in the second semiconductor substrate **117**. It will be appreciated that the source/drain regions **218** may refer to a source or a drain, individually or collectively dependent upon the context. It will also be appreciated that one or more of the source/drain regions **218** may be shared source/drain regions. It will be appreciated that, in some embodiments, the first plurality of transistors **214** may be substantially the same (e.g., comprise similar structural features (and/or have a substantially similar layout)) as the plurality of transistors **124**, or vice versa.

In some embodiments, the first transistor cell **118a** comprises a first isolation structure **220**. The first isolation structure **220** is disposed in the second semiconductor substrate **117**. The first isolation structure **220** laterally surrounds the first plurality of transistors **214**. In some embodiments, the first isolation structure **220** may be, for example, a shallow trench isolation (STI) structure, a deep trench isolation (DTI) structure, or the like. Also shown in the layout view **200** of FIG. 2, the first TSV **136a** is electrically coupled to one of the source/drain regions **218** of the third transistor **214c** (which is illustrated in the layout view **200** of FIG. 2 via a dotted line).

The second chip **104** may also comprise a third transistor cell **118c** of the plurality of transistor cells **118**. In some embodiments, the first transistor cell **118a** neighbors the third transistor cell **118c**. In other embodiments, one or more

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other transistor cells are disposed laterally between the first transistor cell **118a** and the third transistor cell **118c**. In further embodiments, the first transistor cell **118a** and the third transistor cell **118c** are disposed in the first column **122a** of the second plurality of columns **122**. In yet further

embodiments, the first transistor cell **118a** may be spaced further from the third transistor cell **118c** than any other transistor cells that are disposed in the first column **122a**. The third transistor cell **118c** comprises a second plurality of transistors **222**. The second plurality of transistors **222** comprises a first transistor **222a**, a second transistor **222b**, and a third transistor **222c**. In some embodiments, the first transistor **222a** is a reset transistor. In some embodiments, the second transistor **222b** is a source-follower transistor. In some embodiments, the third transistor **222c** is a select transistor (e.g., row select transistor).

The first transistor **222a**, the second transistor **222b**, and the third transistor **222c** each comprise a gate electrode structure **224** overlying a gate dielectric structure (not shown). The gate dielectric structure overlies the second semiconductor substrate **117**. In some embodiments, the gate electrode structure **224** is or comprises, for example, polysilicon, a metal (e.g., aluminum (Al), copper (Cu), titanium (Ti), tantalum (Ta), tungsten (W), molybdenum (Mo), cobalt (Co), or the like), some other conductive material, or a combination of the foregoing. In further embodiments, the gate dielectric structure is or comprises, for example, an oxide (e.g., silicon dioxide (SiO₂)), a high-k dielectric material (e.g., hafnium oxide (HfO), tantalum oxide (TaO), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), aluminum oxide (AlO), zirconium oxide (ZrO), some other dielectric material with a dielectric constant greater than about 3.9), some other dielectric material, or a combination of the foregoing.

The first transistor **222a**, the second transistor **222b**, and the third transistor **222c** each comprise source/drain regions **226** disposed on opposite sides of a corresponding gate electrode structure. For example, the first transistor **222a** comprises source/drain regions **226** disposed on opposite sides of the gate electrode structure of the first transistor **222a**; the second transistor **222b** comprises source/drain regions **226** disposed on opposite sides of the gate electrode structure of the second transistor **222b**; and so forth. The source/drain regions **226** are disposed in the second semiconductor substrate **117**. It will be appreciated that the source/drain regions **226** may refer to a source or a drain, individually or collectively dependent upon the context. It will also be appreciated that one or more of the source/drain regions **226** may be shared source/drain regions. It will be appreciated that, in some embodiments, the second plurality of transistors **222** may be substantially the same (e.g., comprise similar structural features (and/or have a substantially similar layout)) as the plurality of transistors **124**, or vice versa.

In some embodiments, the third transistor cell **118c** comprises a second isolation structure **228**. The second isolation structure **228** is disposed in the second semiconductor substrate **117**. The second isolation structure **228** laterally surrounds the second plurality of transistors **222**. In some embodiments, the second isolation structure **228** may be, for example, a shallow trench isolation (STI) structure, a deep trench isolation (DTI) structure, or the like. Also shown in the layout view **200** of FIG. 2, a third TSV **136c** of the plurality of TSVs **136** is electrically coupled to one of the source/drain regions **226** of the third transistor **222c** (which is illustrated in the layout view **200** of FIG. 2 via a dotted line).

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In some embodiments, the third transistor cell **118c** comprises substantially the same features (e.g., structural features, such as first transistor, second transistor, third transistor, isolation structure, etc.) and/or has substantially the same layout as the first transistor cell **118a**, as shown in the layout view **200** of FIG. 2. In further embodiments, each of the transistor cells of the plurality of transistor cells **118** comprise substantially the same features and/or have substantially the same layout.

Also shown in the layout view **200** of FIG. 2, the third chip **106** comprises the first readout circuit **142a**. The first TSV **136a** is electrically coupled to the first conductive bond structure **143a**. The third TSV **136c** is electrically coupled to a third conductive bond structure **143c** of the first plurality of conductive bond structures **143**. The first conductive bond structure **143a** and the third conductive bond structure **143c** are disposed in the first column **148a**. The third conductive bond structure **143c** and the first conductive bond structure **143a** are electrically coupled to the first readout circuit **142a** (illustrated by solid lines in the layout view **200** of FIG. 2).

In some embodiments, the first readout circuit **142a** comprises an amplifier circuit **232** and an analog-to-digital converter (ADC) **234**. The amplifier circuit **232** comprises one or more readout devices **236** (e.g., transistors) that are disposed on the third semiconductor substrate **140**. The ADC **234** comprises one or more readout devices **238** (e.g., transistors) that are disposed on the third semiconductor substrate **140**. In some embodiments, each of the readout circuits of the first plurality of readout circuits **142** comprise substantially the same features (e.g., structural features, such as the amplifier circuit, the ADC, the one or more readout devices, etc.) and/or have substantially the same layout. In some embodiments, the amplifier circuit **232** comprises substantially the same features (e.g., structural features) as the amplifier circuit **144**, or vice versa. In some embodiments, the ADC **234** comprises substantially the same features (e.g., structural features) as the ADC **146**, or vice versa.

Also shown in the layout view **200** of FIG. 2, a first region **230** of the second chip **104** is disposed laterally between the first transistor cell **118a** and the third transistor cell **118c**. In some embodiments, the first region **230** of the second chip **104** is also disposed laterally between the first TSV **136a** and the third TSV **136c**. In some embodiments, the first region **230** of the second chip **104** laterally separates the first transistor cell **118a** from the third transistor cell **118c**. In some embodiments, the first readout circuit **142a** is disposed within a perimeter of the first region **230** of the second chip **104** (which is illustrated in phantom in the third chip **106** for clarity). In further embodiments, the one or more readout devices **236** and/or the one or more readout devices **238** are disposed within the perimeter of the first region **230** of the second chip **104**. It will be appreciated that each of the other readout circuits of the first plurality of readout circuits **142** (and their corresponding one or more readout devices) may be disposed within a region of the second chip **104** that is substantially similar to the first region **230** of the second chip **104** (e.g., a region of the second chip disposed laterally between two corresponding transistor cells).

FIG. 3 illustrates a cross-sectional view **300** of some embodiments of the high-speed readout image sensor.

As shown in the cross-sectional view **300** of FIG. 3, the first semiconductor substrate **107** has a front side **107f** and a back side **107b** opposite the front side **107f**. In some embodiments, the front side **107f** of the first semiconductor substrate **107** is defined by a first surface (e.g., a front side surface) of the first semiconductor substrate **107**, and the back side **107b** of the first semiconductor substrate **107** is

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defined by a second surface (e.g., a back side surface) of the first semiconductor substrate **107** that is opposite the first surface of the first semiconductor substrate **107**.

In some embodiments, the high-speed readout image sensor is configured to record incident radiation (e.g., photons) that passes through the back side **107b** of the first semiconductor substrate **107** (e.g., backside illumination image sensor). In other embodiments, the image sensor is configured to record incident radiation (e.g., photons) that passes through the front side **107f** of the first semiconductor substrate **107** (e.g., front-side illumination image sensor). The first semiconductor substrate **107** may have a first doping type (e.g., p-type/n-type), or may be intrinsic. In other embodiments, the first semiconductor substrate **107** may have a second doping type (e.g., n-type/p-type) opposite the first doping type.

The one or more photodetectors **202** are disposed in the first semiconductor substrate **107**. In some embodiments, the one or more photodetectors **208** are also disposed in the first semiconductor substrate **107** (see, e.g., FIG. 2). In some embodiments, the one or more photodetectors **202** (and the one or more photodetectors **208**) respectively comprise portions of the first semiconductor substrate **107** having the second doping type. In other embodiments, the one or more photodetectors **202** (and the one or more photodetectors **208**) respectively comprise portions of the first semiconductor substrate **107** having the first doping type. In some embodiments, portions of the first semiconductor substrate **107** adjoining the one or more photodetectors **202** (and the one or more photodetectors **208**) have the first doping type, or may be intrinsic.

The floating diffusion node **206** (and the floating diffusion node **212**) are disposed in the first semiconductor substrate **107**. In some embodiments, the floating diffusion node **206** (and the floating diffusion node **212**) is a region of the first semiconductor substrate **107** having the second doping type. In some embodiments, the one or more transfer gates **204** (and the one or more transfer gates **210**) are disposed along the front side **107f** of the first semiconductor substrate **107**.

Each of the one or more transfer gates **204** (and each of the one or more transfer gates **210**) comprise a gate dielectric structure **302** and a gate electrode structure **304**. In some embodiments, the gate electrode structure **304** is or comprises, for example, polysilicon, a metal (e.g., aluminum (Al), copper (Cu), titanium (Ti), tantalum (Ta), tungsten (W), molybdenum (Mo), cobalt (Co), or the like), some other conductive material, or a combination of the foregoing. In further embodiments, the gate dielectric structure **302** is or comprises, for example, an oxide (e.g., silicon dioxide (SiO₂)), a high-k dielectric material (e.g., hafnium oxide (HfO), tantalum oxide (TaO), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), aluminum oxide (AlO), zirconium oxide (ZrO), some other dielectric material with a dielectric constant greater than about 3.9), some other dielectric material, or a combination of the foregoing. In some embodiments, sidewall spacers **306** are disposed along the front side **107f** of the first semiconductor substrate **107** and laterally surround the one or more transfer gates **204** (and the one or more transfer gates **210**). In some embodiments, the sidewall spacers **306** may be or comprise, for example, an oxide (e.g., SiO₂), a nitride (e.g., silicon nitride (SiN)), an oxy-nitride (e.g., silicon oxynitride (SiO_xN_y)), some other dielectric, or a combination of the foregoing (e.g., oxide-nitride-oxide (ONO) sidewall spacer).

The first chip **102** comprises a first interlayer dielectric (ILD) structure **308**. The first ILD structure **308** is disposed vertically between the first semiconductor substrate **107** and

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the second chip **104**. In some embodiments, the first ILD structure **308** comprises one or more stacked ILD layers, which may respectively comprise a low-k dielectric (e.g., a dielectric material with a dielectric constant less than about 3.9), an oxide (e.g., SiO₂), or the like.

A first interconnect structure **310** (e.g., copper interconnect) is disposed in the first ILD structure **308**. The first interconnect structure **310** is disposed vertically between the first semiconductor substrate **107** and the second chip **104**. The first interconnect structure **310** comprises a first plurality of conductive contacts **310a** (e.g., metal contacts), a first plurality of conductive wires **310b** (e.g., metal wires), a first plurality of conductive vias **310c** (e.g., metal vias), and a second plurality of conductive bond structures **310d** (e.g., metal bond pads). In some embodiments, the first interconnect structure **310** may be or comprise, for example, copper (Cu), aluminum (Al), tungsten (W), gold (Au), some other conductive material, or a combination of the foregoing.

Also shown in the cross-sectional view **300** of FIG. 3, the second semiconductor substrate **117** has a front side **117f** and a back side **117b** opposite the front side **117f**. In some embodiments, the front side **117f** of the second semiconductor substrate **117** is defined by a first surface (e.g., a front side surface) of the second semiconductor substrate **117**, and the back side **117b** of the second semiconductor substrate **117** is defined by a second surface (e.g., a back side surface) of the second semiconductor substrate **117** that is opposite the first surface of the second semiconductor substrate **117**. In some embodiments, the front side **107f** of the first semiconductor substrate **107** is disposed vertically between the back side **107b** of the first semiconductor substrate **107** and the second semiconductor substrate **117**.

The second semiconductor substrate **117** may have the first doping type, or may be intrinsic. In other embodiments, the second semiconductor substrate **117** may have the second doping type. In some embodiments, the second semiconductor substrate **117** may have a same doping type as the first semiconductor substrate **107**. In other embodiments, the second semiconductor substrate **117** may have an opposite doping type as the first semiconductor substrate **107**.

The first plurality of transistors **214** (and the second plurality of transistors **222**) are disposed along the front side **117f** of the second semiconductor substrate **117**. The source/drain regions **218** (and the source/drain regions **226**) are disposed in the second semiconductor substrate **117**. In some embodiments, the source/drain regions **218** (and the source/drain regions **226**) respectively comprise portions of the second semiconductor substrate **117** having the second doping type. In other embodiments, the source/drain regions **218** (and the source/drain regions **226**) respectively comprise portions of the first semiconductor substrate **107** having the first doping type. In some embodiments, portions of the second semiconductor substrate **117** adjoining the source/drain regions **218** (and the source/drain regions **226**) have the first doping type (e.g., p-type/n-type), or may be intrinsic. The first isolation structure **220** (and the second isolation structure **228**) are disposed in the second semiconductor substrate **117**.

In some embodiments, each of the first plurality of transistors **214** (and each of the second plurality of transistors **222**) comprise the gate electrode structure **216** overlying a gate dielectric structure **312**. In some embodiments, the gate dielectric structure **312** is or comprises, for example, an oxide (e.g., silicon dioxide (SiO₂)), a high-k dielectric material (e.g., hafnium oxide (HfO), tantalum oxide (TaO), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), aluminum oxide (AlO), zirconium oxide (ZrO),

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some other dielectric material with a dielectric constant greater than about 3.9), some other dielectric material, or a combination of the foregoing. In further embodiments, sidewall spacers **314** are disposed along the front side **117f** of the second semiconductor substrate **117** and laterally surround the first plurality of transistors **214** (and the second plurality of transistors **222**). In some embodiments, the sidewall spacers **314** may be or comprise, for example, an oxide (e.g., SiO₂), a nitride (e.g., SiN), an oxy-nitride (e.g., SiO_xN_y), some other dielectric, or a combination of the foregoing (e.g., oxide-nitride-oxide (ONO) sidewall spacer).

The second chip **104** comprises a second ILD structure **316**. The second ILD structure **316** is disposed vertically between the first ILD structure **308** and the second semiconductor substrate **117**. In some embodiments, the second ILD structure **316** comprises one or more stacked ILD layers, which may respectively comprise a low-k dielectric (e.g., a dielectric material with a dielectric constant less than about 3.9), an oxide (e.g., SiO₂), or the like.

A second interconnect structure **318** (e.g., copper interconnect) is disposed in the second ILD structure **316**. The second interconnect structure **318** is disposed vertically between the second semiconductor substrate **117** and the first ILD structure **308**. The second interconnect structure **318** comprises a second plurality of conductive contacts **318a** (e.g., metal contacts), a second plurality of conductive wires **318b** (e.g., metal wires), a second plurality of conductive vias **318c** (e.g., metal vias), and a third plurality of conductive bond structures **318d** (e.g., metal bond pads). In some embodiments, the second interconnect structure **318** may be or comprise, for example, copper (Cu), aluminum (Al), tungsten (W), gold (Au), some other conductive material, or a combination of the foregoing.

The second chip **104** is bonded to the first chip **102**. In some embodiments, the first ILD structure **308** is bonded to the second ILD structure **316**. In further embodiments, dielectric-to-dielectric bonds exist at an interface of the first ILD structure **308** and the second ILD structure **316**.

The second interconnect structure **318** is electrically coupled to the first interconnect structure **310**. In some embodiments, the second interconnect structure **318** is electrically coupled to the first interconnect structure **310** via the second plurality of conductive bond structures **310d** and the third plurality of conductive bond structures **318d**. In some embodiments, the second plurality of conductive bond structures **310d** are bonded to the third plurality of conductive bond structures **318d**. In further embodiments, metal-to-metal bonds exist at interfaces in which the second plurality of conductive bond structures **310d** engage the third plurality of conductive bond structures **318d**.

The plurality of TSVs **136** are disposed in the second semiconductor substrate **117**. The plurality of TSVs **136** extend vertically through the second semiconductor substrate **117** from the back side **117b** to the front side **117f**. In some embodiments, the plurality of TSVs **136** are disposed in the second semiconductor substrate **117** and the second ILD structure **316**. In further embodiments, the plurality of TSVs **136** are electrically coupled to the second interconnect structure **318**. In further embodiments, the plurality of TSVs **136** are electrically coupled to one or more of the second plurality of conductive wires **318b**. In some embodiments, the plurality of TSVs **136** may be referred to as back-side through-substrate vias (BTSVs).

In some embodiments, the second chip **104** comprise a dielectric structure **320** disposed along the back side **117b** of the second semiconductor substrate **117**. In some embodiments, the dielectric structure **320** comprises one or more

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stacked dielectric layers, which may respectively comprise a low-k dielectric (e.g., a dielectric material with a dielectric constant less than about 3.9), an oxide (e.g., SiO₂), or the like.

A fourth plurality of conductive bond structures **323** are disposed in the dielectric structure **320**. In some embodiments, a third plurality of conductive vias **322** (e.g., metal vias) are disposed in the dielectric structure **320**. In some embodiments, the third plurality of conductive vias **322** may be or comprise, for example, copper (Cu), aluminum (Al), tungsten (W), gold (Au), some other conductive material, or a combination of the foregoing. In some embodiments, the fourth plurality of conductive bond structures **323** may be or comprise, for example, copper (Cu), aluminum (Al), tungsten (W), gold (Au), some other conductive material, or a combination of the foregoing.

In some embodiments, the third plurality of conductive vias **322** are electrically coupled to the plurality of TSVs **136**, respectively. In further embodiments, the third plurality of conductive vias **322** are electrically coupled to the fourth plurality of conductive bond structures **323**, respectively. In yet further embodiments, the third plurality of conductive vias **322** electrically coupled the plurality of TSVs **136** to the fourth plurality of conductive bond structures **323**, respectively.

Also shown in the cross-sectional view **300** of FIG. **3**, the third semiconductor substrate **140** has a front side **140f** and a back side **140b** opposite the front side **140f**. In some embodiments, the front side **140f** of the third semiconductor substrate **140** is defined by a first surface (e.g., a front side surface) of the third semiconductor substrate **140**, and the back side **140b** of the third semiconductor substrate **140** is defined by a second surface (e.g., a back side surface) of the third semiconductor substrate **140** that is opposite the first surface of the third semiconductor substrate **140**. In some embodiments, the front side **140f** of the third semiconductor substrate **140** is disposed vertically between the back side **140b** of the third semiconductor substrate **140** and the second semiconductor substrate **117**.

The third semiconductor substrate **140** may have the first doping type, or may be intrinsic. In other embodiments, the third semiconductor substrate **140** may have the second doping type. In some embodiments, the third semiconductor substrate **140** may have a same doping type as the second semiconductor substrate **117**. In other embodiments, the third semiconductor substrate **140** may have an opposite doping type as the second semiconductor substrate **117**.

The one or more readout devices **236** and the one or more readout devices **238** are disposed on the third semiconductor substrate **140**. In some embodiments, the one or more readout devices **236** and the one or more readout devices **238** are disposed along the front side **140f** of the third semiconductor substrate **140**.

In some embodiments, the one or more readout devices **236** and the one or more readout devices **238** each comprise a gate dielectric structure **324** and a gate electrode structure **326**. In some embodiments, the gate electrode structure **326** is or comprises, for example, polysilicon, a metal (e.g., aluminum (Al), copper (Cu), titanium (Ti), tantalum (Ta), tungsten (W), molybdenum (Mo), cobalt (Co), or the like), some other conductive material, or a combination of the foregoing. In further embodiments, the gate dielectric structure **324** is or comprises, for example, an oxide (e.g., silicon dioxide (SiO₂)), a high-k dielectric material (e.g., hafnium oxide (HfO), tantalum oxide (TaO), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), aluminum oxide (AlO), zirconium oxide (ZrO), some other dielectric mate-

rial with a dielectric constant greater than about 3.9), some other dielectric material, or a combination of the foregoing. In some embodiments, sidewall spacers **328** are disposed along the front side **140f** of the third semiconductor substrate **140** and laterally surround the gate electrode structures of the one or more readout devices **236** and the gate electrode structures of the one or more readout devices **238**. In some embodiments, the sidewall spacers **328** may be or comprise, for example, an oxide (e.g., SiO_2), a nitride (e.g., SiN), an oxy-nitride (e.g., SiO_xN_y), some other dielectric, or a combination of the foregoing (e.g., oxide-nitride-oxide (ONO) sidewall spacer).

In some embodiments, a third isolation structure **330** is disposed in the third semiconductor substrate **140**. The third isolation structure **330** may laterally surround the one or more readout devices **236** and the one or more readout devices **238**. In some embodiments, the third isolation structure **330** may be, for example, a shallow trench isolation (STI) structure, a deep trench isolation (DTI) structure, or the like.

In some embodiments, a first plurality of doped wells **332** are disposed in the third semiconductor substrate **140**. The first plurality of doped wells **332** are regions of the third semiconductor substrate **140** having the first doping type. In some embodiments, a second plurality of doped wells **334** are disposed in the third semiconductor substrate **140**. The second plurality of doped wells **334** are regions of the third semiconductor substrate **140** having the second doping type.

The third chip **106** comprises a third ILD structure **336**. The third ILD structure **336** is disposed vertically between the second semiconductor substrate **117** and the third semiconductor substrate **140**. In some embodiments, the third ILD structure **336** is disposed vertically between the dielectric structure **320** and the third semiconductor substrate **140**. In some embodiments, the third ILD structure **336** comprises one or more stacked ILD layers, which may respectively comprise a low-k dielectric (e.g., a dielectric material with a dielectric constant less than about 3.9), an oxide (e.g., SiO_2), or the like.

A third interconnect structure **338** (e.g., copper interconnect) is disposed in the third ILD structure **336**. The third interconnect structure **338** is disposed vertically between the second semiconductor substrate **117** and the third semiconductor substrate **140**. In some embodiments, the third interconnect structure **338** is disposed vertically between the dielectric structure **320** and the third semiconductor substrate **140**. The third interconnect structure **338** comprises a third plurality of conductive contacts **338a** (e.g., metal contacts), a third plurality of conductive wires **338b** (e.g., metal wires), a fourth plurality of conductive vias **338c** (e.g., metal vias), and the first plurality of conductive bond structures **143** (e.g., metal bond pads). In some embodiments, the third interconnect structure **338** may be or comprise, for example, copper (Cu), aluminum (Al), tungsten (W), gold (Au), some other conductive material, or a combination of the foregoing.

The third chip **106** is bonded to the second chip **104**. In some embodiments, the dielectric structure **320** is bonded to the third ILD structure **336**. In further embodiments, dielectric-to-dielectric bonds exist at an interface of the dielectric structure **320** and the third ILD structure **336**.

The third interconnect structure **338** is electrically coupled to the second interconnect structure **318**. In some embodiments, the third interconnect structure **338** is electrically coupled to the second interconnect structure **318** via the plurality of TSVs **136**, the fourth plurality of conductive bond structures **323**, and the third plurality of conductive

vias **322**. In some embodiments, the fourth plurality of conductive bond structures **323** are bonded to first plurality of conductive bond structures **143**. In further embodiments, metal-to-metal bonds exist at interfaces in which the fourth plurality of conductive bond structures **323** engage the first plurality of conductive bond structures **143**.

FIG. **4** illustrates a layout diagram **400** of some embodiments of the high-speed readout image sensor.

As shown in the layout diagram **400** of FIG. **4**, the first photodetector cell **108a** may comprise a plurality of photodetector cell units **402**. The plurality of photodetector cell units **402** each comprise a photodetector (see, e.g., one of the one or more photodetectors **202**) and a transfer gate (see, e.g., one of the one or more transfer gates **204**). In some embodiments, the plurality of photodetector cell units **402** are configured to transfer accumulated charges from their photodetectors (via their corresponding transfer gates) to a floating diffusion node (see, e.g., floating diffusion node **206**). In some embodiments, the floating diffusion node of the first photodetector cell **108a** is electrically coupled to (e.g., via the first interconnect structure **310** and the second interconnect structure **318**) the first transistor cell **118a** (and the first plurality of transistors **214**). In further embodiments, the plurality of photodetector cell units **402** are disposed in an array comprising rows and columns.

It will be appreciated that each of the plurality of photodetector cells may comprise a corresponding plurality of photodetector cell units (see, e.g., the plurality of photodetector cell units **402**). In some embodiments, each of the plurality of photodetector cells **108** comprise a same number of photodetector cell units (e.g., 1 photodetector cell unit, 2 photodetector cell units, 16 photodetector cell units, etc.). In some embodiments, the plurality of photodetector cells **108** comprise a same number of individual photodetector cells as the plurality of transistor cells **118** comprises individual transistor cells. In other words, there may be a one-to-one correspondence between the plurality of photodetector cells **108** and the plurality of transistor cells **118**.

FIG. **5** illustrates a layout view **500** of some other embodiments of the high-speed readout image sensor. For clarity in the figures, the layout view **500** of FIG. **5** illustrates a perimeter of the first readout circuit **142a** in phantom (illustrated via a dotted line) on the second chip **104**.

As shown in the layout view **500** of FIG. **5**, the first TSV **136a** is laterally spaced from the first transistor cell **118a** by a first distance **502**. The first distance **502** is greater than or equal to 0.1 micrometers (μm). In some embodiments, if the first distance **502** is less than 0.1 μm , yield of the high-speed readout image sensor may be negatively affected due to a process for forming the plurality of TSVs **136** damaging the first transistor cell **118a** (e.g., unintentionally etching portions of the first transistor cell **118a**). The first TSV **136a** may also be laterally spaced from the third transistor cell **118c** by the first distance **502**.

The third TSV **136c** is laterally spaced from the third transistor cell **118c** by a second distance **504**. The second distance **504** is greater than or equal to 0.1 μm . In some embodiments, if the second distance **504** is less than 0.1 μm , yield of the high-speed readout image sensor may be negatively affected due to a process for forming the plurality of TSVs **136** damaging the third transistor cell **118c** (e.g., unintentionally etching portions of the third transistor cell **118c**). The third TSV **136c** may also be laterally spaced from the first transistor cell **118a** by the second distance **504**.

In some embodiments, the first distance **502** may be substantially the same as the second distance **504** (e.g., substantially the same distance includes small variations due

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to fabrication steps). In other embodiments, the first distance **502** may be different than the second distance **504**. In further embodiments, each of the plurality of TSVs **136** are laterally spaced from their corresponding transistor cell by at least 0.1 μm .

Also shown in the layout view **500** of FIG. **5**, the first isolation structure **220** comprises a sidewall **506**. The second isolation structure **228** comprises a sidewall **508**. The sidewall **506** of the first isolation structure **220** faces the sidewall **508** of the second isolation structure **228**. In some embodiments, the sidewall **506** of the first isolation structure **220** partially defines a perimeter of the first region **230** of the second chip **104**. In further embodiments, the sidewall **508** of the second isolation structure **228** partially defines the perimeter of the first region **230** of the second chip **104**. For example, the sidewall **506** of the first isolation structure **220** may define a first edge/side of the perimeter of the first region **230** of the second chip **104**; and the sidewall **508** of the second isolation structure **228** may define a second edge/side of the perimeter of the first region **230** of the second chip **104**.

The first readout circuit **142a** is laterally spaced from the first transistor cell **118a** by a third distance **510**. The first readout circuit **142a** is laterally spaced from the third transistor cell **118c** by a fourth distance **512**. In some embodiments, the third distance **510** is different than the fourth distance **512**. In other embodiments, the third distance **510** is substantially the same as the fourth distance **512**. In some embodiments, because the third distance **510** is substantially the same as the fourth distance **512**, the conductive paths that electrically couple the transistor cells of the plurality of transistor cells **118** that are disposed in the first column **122a** to the first readout circuit **142a** may be further reduced. For example, in some embodiments, the first transistor cell **118a** may be spaced further from the third transistor cell **118c** than any other of the transistor cells of the plurality of transistor cells **118** that are disposed in the first column **122a**. Thus, if the third distance **510** is substantially the same as the fourth distance **512**, the first readout circuit **142a** will be disposed along a centerline of the first array, thereby further reducing the conductive paths that electrically couple the transistor cells (of the first column **122a**) to the first readout circuit **142a**.

FIG. **6** illustrates a schematic diagram **600** of some embodiments of a third chip **106** of some embodiments of the high-speed readout image sensor.

As shown in the schematic diagram **600** of FIG. **6**, in some embodiments, each of the conductive bond structures of the first plurality of conductive bond structures **143** that are disposed in the first column **148a** are electrically coupled to the first readout circuit **142a** (illustrated via solid lines). In further embodiments, each of the conductive bond structures of the first plurality of conductive bond structures **143** that are disposed in the second column **148b** are electrically coupled to the second readout circuit **142b** (illustrated via solid lines). In further embodiments, each of the first plurality of readout circuits **142** may comprise an amplifier circuit **144** and an ADC **146**. It will be appreciated that the third chip **106** may comprise additional circuits that are substantially the same as those illustrated in the schematic diagram **600** of FIG. **6** (e.g., other readout circuits that are electrically coupled to other conductive bond pads that are disposed in other columns). In some embodiments, if a high-speed readout image sensor comprises the circuits illustrated in the schematic diagram **600** of FIG. **6**, the first plurality of readout circuits **142** may be referred to as column-level readout circuits (e.g., because the first plurality

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of readout circuits **142** are configured to receive (and read out) signals from a column of photodetector cells).

FIG. **7** illustrates a layout diagram **700** of some other embodiments of the high-speed readout image sensor.

As shown in the layout diagram **700** of FIG. **7**, in some embodiments, the first plurality of conductive bond structures **143** are electrically coupled to the first plurality of readout circuits **142**, respectively. For example, the first conductive bond structure **143a** is electrically coupled to the first readout circuit **142a**; the second conductive bond structure **143b** is electrically coupled to the second readout circuit **142b**; and so forth. It will be appreciated that the ellipsis symbols (...) shown in the layout diagram **700** of FIG. **7** illustrate that the first chip **102**, the second chip **104**, and/or the third chip **106** may be larger than illustrated and may comprise additional iterations of the features (e.g., additional TSVs, additional photodetectors cells, additional transistors cells, etc.) illustrated in the layout diagram **700** of FIG. **7**.

Also shown in the layout diagram **700** of FIG. **7**, a fourth conductive bond structure **143d** of the first plurality of conductive bond structures **143** is electrically coupled to a fourth readout circuit **142d** of the first plurality of readout circuits **142**. The fourth conductive bond structure **143d** is electrically coupled to a fourth TSV **136d** of the plurality of TSVs **136**. The fourth TSV **136d** of the plurality of TSVs **136** is electrically coupled to a fourth transistor cell **118d** of the plurality of transistor cells **118**.

A fifth conductive bond structure **143e** of the first plurality of conductive bond structures **143** is electrically coupled to a fifth readout circuit **142e** of the first plurality of readout circuits **142**. The fifth conductive bond structure **143e** is electrically coupled to a fifth TSV **136e** of the plurality of TSVs **136**. The fifth TSV **136e** of the plurality of TSVs **136** is electrically coupled to a fifth transistor cell **118e** of the plurality of transistor cells **118**. In some embodiments, each of the first transistor cell **118a**, the second transistor cell **118b**, the fourth transistor cell **118d**, and the fifth transistor cell **118e** are disposed in the first column **122a**. In further embodiments, the fourth readout circuit **142d** and the fifth readout circuit **142e** are both disposed laterally between the first transistor cell **118a** and the second transistor cell **118b**.

FIG. **8** illustrates a schematic diagram **800** of some embodiments of a third chip **106** of some embodiments of the high-speed readout image sensor of FIG. **7**.

As shown in the schematic diagram **800** of FIG. **8**, in some embodiments, the first plurality of conductive bond structures **143** are electrically coupled to the first plurality of readout circuits **142**, respectively (illustrated via solid lines). In further embodiments, each of the first plurality of readout circuits **142** may comprise an amplifier circuit **144** and an ADC **146**. It will be appreciated that the third chip **106** may comprise additional circuits that are substantially the same as those illustrated in the schematic diagram **800** of FIG. **8** (e.g., other readout circuits that are respectively electrically coupled to other conductive bond pads). In some embodiments, if a high-speed readout image sensor comprises the circuits illustrated in the schematic diagram **800** of FIG. **8**, the first plurality of readout circuits **142** may be referred to as pixel-level readout circuits (e.g., because each of the first plurality of readout circuits **142** are configured to receive (and read out) signals from one of the plurality of photodetector cells **108**). In some embodiments, because the first plurality of conductive bond structures **143** are respectively electrically coupled to the first plurality of readout circuits **142**, the high-speed readout image sensor may have an even

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smaller RC delay (e.g., due to the first plurality of readout circuits **142** processing signals at a pixel level).

FIG. **9** illustrates a layout diagram **900** of some other embodiments of the high-speed readout image sensor.

As shown in the layout diagram **900** of FIG. **9**, in some embodiments, the second chip **104** may comprise the first plurality of readout circuits **142**. In some embodiments, the one or more readout devices (see, e.g., the one or more readout devices **236**) are disposed on the second semiconductor substrate **117** (e.g., along the front side **117f** of the second semiconductor substrate **117**). The first plurality of readout circuits **142** are electrically coupled to the plurality of transistor cells **118**, respectively. In some embodiments, the first plurality of readout circuits **142** are respectively electrically coupled to the plurality of transistor cells **118** via conductive features of the second interconnect structure **318** (e.g., the second plurality of conductive contacts **318a**, the second plurality of conductive wires **318b**, the second plurality of conductive vias **318c**, etc.). In some embodiments, the second interconnect structure **318** may also electrically coupled together TSVs disposed in a column of TSVs (illustrated by solid lines).

Also shown in the layout diagram **900** of FIG. **9**, in some embodiments, the third chip **106** comprises a second plurality of readout circuits **902**. In some embodiments, each of the second plurality of readout circuits **902** comprise one or more image processing circuits (e.g., anti-aliasing circuit, smoothing circuit, low pass filter, high pass filter, compression circuitry, some other image processing circuit, or a combination of the foregoing). In further embodiments, each of the second plurality of readout circuits **902** comprise substantially the same features and/or have substantially the same layout.

The first plurality of conductive bond structures **143** are electrically coupled to the second plurality of readout circuits **902**. More specifically, in some embodiments, the conductive bond structures of the first plurality of conductive bond structures **143** of a given column of the third plurality of columns **148** are electrically coupled to a corresponding one of the second plurality of readout circuits **902**. For example, in some embodiments, each of the first plurality of conductive bond structures **143** that are disposed in the first column **148a** are electrically coupled to a first readout circuit **902a** of the second plurality of readout circuits **902**; each of the first plurality of conductive bond structures **143** that are disposed in the second column **148b** are electrically coupled to a second readout circuit **902b** of the second plurality of readout circuits **902**; and so forth. It will be appreciated that the ellipsis symbols (. . .) shown in the layout diagram **900** of FIG. **9** illustrate that the first chip **102**, the second chip **104**, and/or the third chip **106** may be larger than illustrated and may comprise additional iterations of the features (e.g., additional TSVs, additional photodetectors cells, additional transistors cells, additional readout circuits, etc.) illustrated in the layout diagram **900** of FIG. **9**.

In some embodiments, the first plurality of readout circuits **142** are configured to receive first electrical signals from the plurality of transistor cells **118** and output second electrical signals, which are based on the first electrical signals, to the plurality of TSVs **136**. For example, in some embodiments, the first readout circuit **142a** is configured to receive a first electrical signal from the first transistor cell **118a** and output a second electrical signal, which is based on the first electrical signal, to the first TSV **136a**; the second readout circuit **142b** is configured to receive a third electrical signal from the second transistor cell **118b** and output a

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fourth electrical signal, which is based on the third electrical signal, to the second TSV **136b**; and so forth. In further embodiments, the first electrical signals correspond to a number of charges accumulated in the one or more photodetectors of the plurality of photodetector cells **108**.

In some embodiments, a readout circuit of the first plurality of readout circuits **142** corresponds to a readout circuit of the second plurality of readout circuits **902** if such readout circuit of the first plurality of readout circuits **142** is electrically coupled to such readout circuit of the second plurality of readout circuits **902** via one of the first plurality of conductive bond structures **143**. For example, in some embodiments, each of the conductive bond structures of the first plurality of conductive bond structures **143** that are disposed in the first column **148a** are electrically coupled to the first readout circuit **902a**; and thus each of the first plurality of readout circuits **142** that are electrically coupled to the conductive bond structures of the first plurality of conductive bond structures **143** that are disposed in the first column **148a** correspond to the first readout circuit **902a**. In further embodiments, the second plurality of readout circuits **902** are configured to receive the second electrical signals from their corresponding readout circuits of the first plurality of readout circuits **142** and output third electrical signals (e.g., image processed signals). For example, in some embodiments, the first readout circuit **902a** is configured to receive the second electrical signal from the first readout circuit **142a** and output a fifth electrical signal that is based on the second electrical signal; the first readout circuit **902a** is configured to receive the fourth electrical signal from the second readout circuit **142b** and output a sixth electrical signal that is based on the fourth electrical signal; and so forth.

Also shown in the layout diagram **900** of FIG. **9**, a second region **904** of the second chip **104** is disposed laterally between the first transistor cell **118a** and the fifth transistor cell **118e**. It will also be appreciated that, in some embodiments, the second region **904** of the second chip **104** may be substantially the same (e.g., comprise similar structural features (and/or have a substantially similar layout)) as the first region **230** of the second chip **104**. In some embodiments, the first readout circuit **142a** is disposed within a perimeter of the second region **904** of the second chip **104**. In further embodiments, the first TSV **136a** may be disposed in the perimeter of the second region **904** of the second chip **104**.

A third region **906** of the second chip **104** is disposed laterally between the fifth transistor cell **118e** and the fourth transistor cell **118d**. In some embodiments, the third region **906** of the second chip **104** is also disposed laterally between the first TSV **136a** and the fourth TSV **136d**. It will also be appreciated that, in some embodiments, the third region **906** of the second chip **104** may be substantially the same (e.g., comprise similar structural features (and/or have a substantially similar layout)) as the first region **230** of the second chip **104**. In some embodiments, the fifth readout circuit **142e** is disposed within a perimeter of the third region **906** of the second chip **104**. In further embodiments, the fifth TSV **136e** may be disposed in the perimeter of the third region **906** of the second chip **104**. In some embodiments, because the second chip comprises the first plurality of readout circuits **142**, the third chip **106** may comprise additional image processing circuits (see, e.g., the second plurality of readout circuits **902**), thereby increasing the applications in which the high-speed readout image sensor may be employed.

In some embodiments (not shown), the first plurality of readout circuits **142** may respectively have a part that is disposed within the second chip **104** and a part that is disposed in the third chip **106**. For example, a first readout circuit of the first plurality of readout circuits **142** may have a part in the second chip **104** and a part in the third chip **106**, a second readout circuit of the first plurality of readout circuits **142** may have a part in the second chip **104** and a part in the third chip **106**, etc. By having the first plurality of readout circuits **142** split between the second chip **104** and the third chip **106**, an area of the third chip **106** consumed by the first plurality of readout circuits **142** can be reduced thereby allowing for more area for image signal processing functionality within the third chip. In some embodiments, the third chip **106** may be formed at a smaller processing node than the second chip **104**, such that the integrated chip structure can on the whole be made to be more cost effective by splitting the first plurality of readout circuits **142** between the second chip **104** and the third chip **106**. FIGS. **10A-10B** through **14A-14B** illustrate a series of various views of some embodiments of a method for forming a high-speed readout image sensor. Figures with a suffix “A” (e.g., FIG. **10A**) illustrate a series of cross-sectional views **1000a-1400a** of some embodiments of a method for forming a high-speed readout image sensor. Figures with a suffix “B” (e.g., FIG. **10B**) illustrate a series of layout views **1000b-1400b** of corresponding figures of FIGS. **10A-14A** (e.g., FIG. **10B** illustrates a layout view of some embodiments of the structure illustrated in FIG. **10A**).

As shown in the various views **1000a-1000b** of FIGS. **10A-10B**, a workpiece **1002** is received. The workpiece **1002** comprises a second semiconductor substrate **117**. The second semiconductor substrate **117** comprises a front side **117f** and a back side **117b**. A plurality of transistor cells **118** are disposed on the second semiconductor substrate **117**. In some embodiments, the plurality of transistor cells **118** are laterally spaced.

A second ILD structure **316** is disposed along the front side **117f** of the second semiconductor substrate **117**. A second interconnect structure **318** is disposed in the second ILD structure **316**. The second interconnect structure **318** comprises a second plurality of conductive contacts **318a** (e.g., metal contacts), a second plurality of conductive wires **318b** (e.g., metal wires), a second plurality of conductive vias **318c** (e.g., metal vias), and a third plurality of conductive bond structures **318d** (e.g., metal bond pads).

In some embodiments, the second plurality of conductive wires **318b** comprises a first conductive wire **318b₁**. In further embodiments, the first conductive wire **318b₁** is electrically coupled to the transistor cells of the plurality of transistor cells **118** of a given column of the second plurality of columns **122**. For example, a first transistor cell **118a**, a second transistor cell **118b**, and a third transistor cell **118c** are each disposed in a first column **122a** of a second plurality of columns **122**; and the first conductive wire **318b₁** is electrically coupled to each of the first transistor cell **118a**, the second transistor cell **118b**, and the third transistor cell **118c**. In further embodiments, the first conductive wire **318b₁** is electrically coupled to a source/drain region of a third transistor (see, e.g., the third transistor **214c**, the third transistor **222c**, the third transistor **124c**, etc.) of each of the transistor cells of the plurality of transistor cells **118** that are disposed in the first column **122a**. In yet further embodiments, the first conductive wire **318b₁** may be disposed in a metal layer (e.g., metal **1**) of the second interconnect struc-

ture **318** that is disposed nearer the second semiconductor substrate **117** than any other metal layer of the second interconnect structure **318**.

In some embodiments, the workpiece **1002** also comprises a first chip **102**. In other embodiments, the workpiece **1002** may not comprise the first chip **102** (e.g., the first chip **102** is bonded to the second chip **104** at a later stage in fabrication). The first chip **102** comprises a first semiconductor substrate **107**. The first semiconductor substrate **107** has a front side **107f** and a back side **107b**. A plurality of photodetector cells **108** are disposed in/on the first semiconductor substrate **107**.

A first ILD structure **308** is disposed along the front side **107f** of the first semiconductor substrate **107**. The first ILD structure **308** is disposed between the second ILD structure **316** and the front side **107f** of the first semiconductor substrate **107**. A first interconnect structure **310** is disposed in the first ILD structure **308**. The first interconnect structure **310** comprises a first plurality of conductive contacts **310a** (e.g., metal contacts), a first plurality of conductive wires **310b** (e.g., metal wires), a first plurality of conductive vias **310c** (e.g., metal vias), and a second plurality of conductive bond structures **310d** (e.g., metal bond pads). In some embodiments, the workpiece **1002** is formed by known complementary metal-oxide-semiconductor (CMOS) processes. Further, it will be appreciated that the features (e.g., structural features) of the workpiece **1002** may be substantially the same and/or have substantially the same layout as corresponding features described above.

As shown in the various views **1100a-1100b** of FIGS. **11A-11B**, a plurality of through-substrate via (TSV) openings **1102** are formed in the second semiconductor substrate **117**. In some embodiments, the plurality of TSV openings **1102** are also formed in the second ILD structure **316**. The plurality of TSV openings **1102** are formed extending vertically through the second semiconductor substrate **117**. In some embodiments, the plurality of TSV openings **1102** are formed extending vertically into the second ILD structure **316**.

In some embodiments, the plurality of TSV openings **1102** respectively expose portions of the first conductive wire **318b₁**. For example, the plurality of TSV openings **1102** comprises a first TSV opening **1102a**, a second TSV opening **1102b**, a third TSV opening **1102c**, and so forth. The first TSV opening **1102a** exposes a first portion of the first conductive wire **318b₁**; the second TSV opening **1102b** exposes a second portion of the first conductive wire **318b₁**; the third TSV opening **1102c** exposes a third portion of the first conductive wire **318b₁**; and so forth. In some embodiments, the third TSV opening **1102c** is formed laterally between the first transistor cell **118a** and the second transistor cell **118b**.

In some embodiments, a process for forming the plurality of TSV openings **1102** comprises forming a patterned masking layer (not shown) (e.g., positive/negative photoresist, a hardmask, etc.) along/over the back side **117b** of the second semiconductor substrate **117**. The patterned masking layer may be formed by forming a masking layer (not shown) on/over the back side **117b** of the second semiconductor substrate **117** (e.g., via a spin-on process), exposing the masking layer to a pattern (e.g., via a lithography process, such as photolithography, extreme ultraviolet lithography, or the like), and developing the masking layer to form the patterned masking layer. Thereafter, with the patterned masking layer in place, an etching process is performed on the second semiconductor substrate **117** and the second ILD structure **316** to selectively etch the second semiconductor

substrate **117** and the second ILD structure **316** according to the patterned masking layer. The etching process removes unmasked portions of the second semiconductor substrate **117** and unmasked portions of the second ILD structure **316**, thereby forming the plurality of TSV openings **1102**. In some embodiments, the etching process may be or comprise, for example, a wet etching process, a dry etching process, a reactive ion etching (RIE) process, some other etching process, or a combination of the foregoing. In further embodiments, the etching process stops on the first conductive wire **318b₁** (e.g., the first conductive wire **318b₁** acts as an etch stop layer during the etching process).

It will be appreciated that the various views **1100a-1100b** of FIGS. **11A-11B** only illustrate the formation of some of the TSV openings of the plurality of TSV openings **1102**. Thus, it will also be appreciated that other TSV openings of the plurality of TSV openings **1102** may also be formed in the second semiconductor substrate **117** (and the second ILD structure **316**). Moreover, it will be appreciated that the other TSV openings of the plurality of TSV openings **1102** may be formed so that they expose portions of other conductive wires of the second plurality of conductive wires **318b**. For example, a first group of TSV openings (a portion of which is illustrated in the various views **1100a-1100b** of FIGS. **11A-11B**) are formed to expose portions of the first conductive wire **318b₁**; a second group of TSV openings (not shown) are formed to expose a second conductive wire (not shown) of the second plurality of conductive wires **318b**; a third group of TSV openings (not shown) are formed to expose a third conductive wire (not shown) of the second plurality of conductive wires **318b**; and so forth.

As shown in the various views **1200a-1200b** of FIGS. **12A-12B**, a plurality of TSVs **136** are respectively formed in the plurality of TSV openings **1102** (see, e.g., FIGS. **11A-11B**). In some embodiments, the plurality of TSVs **136** are formed electrically coupled to a corresponding conductive wire of the second plurality of conductive wires **318b** (e.g., a first group of TSVs are formed electrically coupled to the first conductive wire **318b₁**; a second group of TSVs are formed electrically coupled to the second conductive wire of the second plurality of conductive wires **318b**; and so forth).

In some embodiments, the plurality of TSVs **136** comprises a first TSV **136a**, a second TSV **136b**, and a third TSV **136c**. In further embodiments, the first TSV **136a** is formed in the first TSV opening **1102a**; the second TSV **136b** is formed in the second TSV opening **1102b**; and the third TSV **136c** is formed in the third TSV opening **1102c**. In yet further embodiments, each of the first TSV **136a**, the second TSV **136b**, and the third TSV **136c** are formed electrically coupled to the first conductive wire **318b₁**.

Also shown in the various views **1200a-1200b** of FIGS. **12A-12B**, a plurality of dielectric liner structures **1202** may be respectively formed in the plurality of TSV openings **1102**. In other embodiments, the plurality of dielectric liner structures **1202** may be omitted. The plurality of dielectric liner structures **1202** are formed along sidewalls of the plurality of TSV openings **1102**, respectively. In some embodiments, the plurality of dielectric liner structures **1202** are formed along sidewalls of the second semiconductor substrate **117**. In further embodiments, the plurality of dielectric liner structures **1202** may be formed along sidewalls of the second ILD structure **316**. In some embodiments, the plurality of dielectric liner structures **1202** may be or comprise, for example, an oxide (e.g., SiO₂), a nitride (e.g., SiN), an oxy-nitride (e.g., SiON), a carbide (e.g., silicon carbide (SiC)), some other dielectric material, or a combination of the foregoing. In some embodiments, a

process for forming the plurality of dielectric liner structures **1202** comprises depositing or growing the dielectric liner structures **1202** along the sidewalls of the plurality of TSV openings **1102**. The dielectric liner structures **1202** may be deposited or grown by, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), thermal oxidation, some other deposition process, or a combination of the foregoing.

In some embodiments, a process for forming the plurality of TSVs **136** comprises depositing a conductive material in the plurality of TSV openings **1102**, thereby forming the plurality of TSVs **136** in the plurality of TSV openings **1102**. In some embodiments, the conductive material is or comprises, for example, copper (Cu), aluminum (Al), gold (Au), silver (Ag), platinum (Pt), polysilicon, some other conductive material, or a combination of the foregoing. The conductive material may be deposited by, for example, CVD, PVD, ALD, sputtering, electrochemical plating, electroless plating, some other deposition process, or a combination of the foregoing. In some embodiments, a planarization process (e.g., chemical-mechanical polishing (CMP)) is performed on the conductive material to planarize surfaces of the TSVs **136** with the back side **117b** of the second semiconductor substrate **117**.

As shown in the various views **1300a-1300b** of FIGS. **13A-13B**, a dielectric structure **320** is formed along the back side **117b** of the second semiconductor substrate **117**. In some embodiments, the dielectric structure **320** is formed on the back side **117b** of the second semiconductor substrate **117**, on the plurality of TSVs **136**, and/or on the plurality of dielectric liner structures **1202**.

Also shown in the various views **1300a-1300b** of FIGS. **13A-13B**, a third plurality of conductive vias **322** are formed in the dielectric structure **320**. The third plurality of conductive vias **322** are formed electrically coupled to the plurality of TSVs **136**, respectively. For example, a first conductive via **322a** of the third plurality of conductive vias **322** is formed electrically coupled to the first TSV **136a**; a second conductive via (not shown) of the third plurality of conductive vias **322** is formed electrically coupled to the second TSV **136b**; a third conductive via (not shown) of the third plurality of conductive vias **322** is formed electrically coupled to the third TSV **136c**; and so forth.

Also shown in the various views **1300a-1300b** of FIGS. **13A-13B**, a fourth plurality of conductive bond structures **323** are formed in the dielectric structure **320**. The fourth plurality of conductive bond structures **323** are formed electrically coupled to the third plurality of conductive vias **322**, respectively. For example, a first conductive bond structure **323a** of the fourth plurality of conductive bond structures **323** is formed electrically coupled to the first conductive via **322a** of the third plurality of conductive vias **322**; a second conductive bond structure **323b** of the fourth plurality of conductive bond structures **323** is formed electrically coupled to the second conductive via of the third plurality of conductive vias **322**; a third conductive bond structure **323c** of the fourth plurality of conductive bond structures **323** is formed electrically coupled to the third conductive via of the third plurality of conductive vias **322**; and so forth.

In some embodiments, a process for forming the dielectric structure **320**, the third plurality of conductive vias **322**, and the fourth plurality of conductive bond structures **323** comprises forming a first dielectric layer on the back side **117b** of the second semiconductor substrate **117**, on the plurality of TSVs **136**, and on the plurality of dielectric liner structures **1202**. The first dielectric layer may be or comprise, for

example, a low-k dielectric (e.g., a dielectric material with a dielectric constant less than about 3.9), an oxide (e.g., SiO₂), or the like. The first dielectric layer may be formed by, for example, CVD, PVD, ALD, a spin-on process, some other deposition process, or a combination of the foregoing.

Thereafter, a patterned masking layer (e.g., positive/negative photoresist, hardmask, etc.) is formed on the first dielectric layer. With the patterned masking layer in place, an etching process (e.g., a wet etching process, dry etching process, RIE process, or the like) is performed on the first dielectric layer, thereby forming a first plurality of openings (not shown) in the first dielectric layer according to the patterned masking layer. Subsequently, the patterned masking layer is stripped away (e.g., via a plasma ashing process). A metal material is then deposited on the first dielectric layer and in the first plurality of openings. The metal material may be or comprise, for example, copper (Cu), aluminum (Al), tungsten (W), gold (Au), some other conductive material, or a combination of the foregoing. The metal material may be deposited by, for example, CVD, PVD, ALD, sputtering, electrochemical plating, electroless plating, some other deposition process, or a combination of the foregoing. Thereafter, a planarization process (e.g., CMP) is performed on the metal material to remove a portion of the metal material, thereby leaving remaining portions of the metal material in the first plurality of openings as the third plurality of conductive vias **322**.

Thereafter, a second dielectric layer is formed on the first dielectric layer and on the third plurality of conductive vias **322**. The second dielectric layer may be or comprise, for example, a low-k dielectric (e.g., a dielectric material with a dielectric constant less than about 3.9), an oxide (e.g., SiO₂), or the like. The second dielectric layer may be formed by, for example, CVD, PVD, ALD, a spin-on process, some other deposition process, or a combination of the foregoing.

Thereafter, a patterned masking layer (e.g., positive/negative photoresist, hardmask, etc.) is formed on the second dielectric layer. With the patterned masking layer in place, an etching process (e.g., a wet etching process, dry etching process, RIE process, or the like) is performed on the second dielectric layer, thereby forming a second plurality of openings (not shown) in the second dielectric layer according to the patterned masking layer. Subsequently, the patterned masking layer is stripped away (e.g., via a plasma ashing process). A metal material is then deposited on the second dielectric layer and in the second plurality of openings. The metal material may be or comprise, for example, copper (Cu), aluminum (Al), tungsten (W), gold (Au), some other conductive material, or a combination of the foregoing. The metal material may be deposited by, for example, CVD, PVD, ALD, sputtering, electrochemical plating, electroless plating, some other deposition process, or a combination of the foregoing. Thereafter, a planarization process (e.g., CMP) is performed on the metal material to remove a portion of the metal material, thereby leaving remaining portions of the metal material in the second plurality of openings as the fourth plurality of conductive bond structures **323**. In some embodiments, forming the fourth plurality of conductive bond structures **323** completes formation of the dielectric structure **320**, which may comprise the first dielectric layer and the second dielectric layer. In further embodiments, forming the fourth plurality of conductive bond structures **323** and the dielectric structure **320** completes formation of a second chip **104**.

As shown in the various views **1400a-1400b** of FIGS. **14A-14B**, a third chip **106** is bonded to the second chip **104**. The third chip **106** comprises a third semiconductor sub-

strate **140**. The third semiconductor substrate **140** has a front side **140f** and a back side **140b**. In some embodiments, the third chip **106** comprises a first plurality of readout circuits **142**. In other embodiments, the third chip **106** comprises a second plurality of readout circuits **902**. In further embodiments, each of the first plurality of readout circuits **142** may comprise an amplifier circuit and/or an ADC (see, e.g., FIG. **1**). The third chip **106** also comprises a third ILD structure **336** disposed along the front side **140f** of the third semiconductor substrate **140**. A third interconnect structure **338** is disposed in the third ILD structure **336**. The third interconnect structure **338** comprises a third plurality of conductive contacts **338a**, a third plurality of conductive wires **338b**, a fourth plurality of conductive vias **338c**, and the first plurality of conductive bond structures **143**.

In some embodiments, a process for bonding the third chip **106** to the second chip **104** comprises bonding the first plurality of conductive bond structures **143** respectively to the fourth plurality of conductive bond structures **323**. For example, the first conductive bond structure **143a** is bonded to the first conductive bond structure **323a**; the second conductive bond structure **143b** is bonded to the second conductive bond structure **323b**; and the third conductive bond structure **143c** is bonded to the third conductive bond structure **323c**; and so forth. In further embodiments, the process for bonding the third chip **106** to the second chip **104** may comprise bonding the third ILD structure **336** to the dielectric structure **320**.

In some embodiments, the third chip **106** is bonded to the second chip **104** so that the second semiconductor substrate **117** is disposed vertically between the third ILD structure **336** and the second ILD structure **316**. In some embodiments, the third chip **106** is bonded to the second chip **104** so that the first plurality of conductive bond structures **143** are respectively electrically coupled to the fourth plurality of conductive bond structures **323**. In yet further embodiments, the third chip **106** may be bonded to the second chip **104** by, for example, a dielectric-to-dielectric and a metal-to-metal bonding process (e.g., a bonding process that forms one or more dielectric-to-dielectric interfaces and one or more metal-to-metal interfaces), a eutectic bonding process, or some other bonding process.

For clarity, it will be appreciated that spatially relative terms (e.g., over, under, upper, lower, etc.) used herein to describe the structures illustrated in the figures are generally based on the orientation of such structures as illustrated in their respective figures. For example, in describing the structure illustrated in FIG. **14**, it may be said that the third chip **106** overlies the second chip **104**. On the other hand, in describing the structure illustrated in FIG. **3**, it may be said that the second chip **104** overlies the third chip **106**.

FIG. **15** illustrates a flowchart **1500** of some embodiments of a method for forming a high-speed readout image sensor. While the flowchart **1500** of FIG. **15** is illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

At act **1502**, a first chip is formed. FIGS. **10A-10B** through FIGS. **13A-13B** illustrate a series of various views

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of some embodiments corresponding to act 1502. In some embodiments, act 1502 comprises act 1504 and act 1506.

At act 1504, a workpiece is received. The workpiece comprises a plurality of transistor cells that are disposed on a first semiconductor substrate. FIGS. 10A-10B illustrate various views 1000a-1000b of some embodiments corresponding to act 1504.

At act 1506, a plurality of through-substrate vias (TSVs) are formed in the first semiconductor substrate. The plurality of TSVs are formed respectively electrically coupled to the plurality of transistor cells. FIGS. 11A-11B through FIGS. 12A-12B illustrate a series of various views of some embodiments corresponding to act 1506.

At act 1508, a second chip is bonded to the first chip. The second chip comprises one or more readout circuits disposed on a second semiconductor substrate. The second chip is bonded to the first chip so that the plurality of transistor cells are electrically coupled to the one or more readout circuits via the plurality of TSVs. FIGS. 14A-14B illustrate various views 1400a-1400b of some embodiments corresponding to act 1508.

In some embodiments, the present application provides an image sensor. The image sensor comprises a first chip. The first chip comprises a first semiconductor substrate, wherein the first semiconductor substrate has a first side and a second side, and wherein the second side of the first semiconductor substrate is opposite the first side of the first semiconductor substrate. The first chip comprises a first transistor cell and a second transistor cell that are disposed in a transistor cell array, wherein the first transistor cell comprises a first plurality of transistors disposed along the first side of the first semiconductor substrate, wherein the first transistor cell is configured to operate a first photodetector cell, wherein the second transistor cell comprises a second plurality of transistors disposed along the first side of the first semiconductor substrate, wherein the second transistor cell is configured to operate a second photodetector cell, and wherein the second transistor cell is laterally spaced from the first transistor cell. The first chip comprises a first through-substrate via (TSV) that extends vertically through the first semiconductor substrate, wherein the first transistor cell is electrically coupled to the first TSV, wherein the first transistor cell is configured to provide a first signal to the first TSV that corresponds to a number of charges accumulated in a photodetector of the first photodetector cell. The first chip comprises a second TSV that extends vertically through the first semiconductor substrate, wherein the second transistor cell is electrically coupled to the second TSV, wherein the second transistor cell is configured to provide a second signal to the second TSV that corresponds to a number of charges accumulated in a photodetector of the second photodetector cell. The image sensor comprises a second chip. The second chip comprises a second semiconductor substrate, wherein the second chip is bonded to the first chip. The second chip comprises a first interlayer dielectric (ILD) structure that is disposed vertically between the second semiconductor substrate and first semiconductor substrate, and wherein the first side of the first semiconductor substrate is disposed vertically between the second side of the first semiconductor substrate and the second semiconductor substrate. The second chip comprises a first readout circuit that is electrically coupled to the first TSV and to the second TSV, wherein the first readout circuit is disposed laterally, at least partially, between the first TSV and the second TSV.

In some embodiments, the first readout circuit comprises a first part that is within the second chip and a second part that is within a third chip bonded to the second chip.

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In some embodiments, the first readout circuit comprises an amplifier circuit comprising one or more first readout devices disposed on the second semiconductor substrate.

In further embodiments, the one or more first readout devices are disposed laterally between the first TSV and the second TSV.

In some embodiments, the first transistor cell comprises a first select transistor; a source/drain region of the first select transistor is electrically coupled to the first TSV; the second transistor cell comprises a second select transistor; and a source/drain region of the second select transistor is electrically coupled to the second TSV.

In some embodiments, the first TSV is laterally spaced from the first transistor cell and the second transistor cell by at least about 0.1 micrometers; and the second TSV is laterally spaced from the first transistor cell and the second transistor cell by at least about 0.1 micrometers.

In some embodiments, the image sensor further comprises a third chip. The third chip comprises a third semiconductor substrate, wherein the third chip is bonded to the first chip, wherein the photodetector of the first photodetector cell and the photodetector of the second photodetector cell are both disposed in the third semiconductor substrate, and wherein the first chip is disposed vertically between the second chip and the third chip.

In some embodiments, the transistor cell array comprises a plurality of transistor cells; the transistor cell array comprises a plurality of rows and a plurality of columns; the first transistor cell is one transistor cell of the plurality of transistor cells; the second transistor cell is another transistor cell of the plurality of transistor cells; and both the first transistor cell and the second transistor cell are disposed in a first column of the plurality of columns.

In further embodiments, the first transistor cell neighbors the second transistor cell.

In further embodiments, the first transistor cell is spaced further from the second transistor cell than any other transistor cell of the plurality of transistor cells that are disposed in the first column.

In further embodiments, the first readout circuit is spaced laterally from the first transistor cell by a first distance; the first readout circuit is spaced laterally from the second transistor cell by a second distance; and the first distance is about the same as the second distance.

In some embodiments, the second chip comprises a second readout circuit electrically coupled to the second TSV; the second readout circuit is configured to receive the second signal; and the second readout circuit is laterally spaced from the first readout circuit.

In further embodiments, the transistor cell array comprises a plurality of transistor cells; the transistor cell array comprises a plurality of rows and a plurality of columns; the first transistor cell is one transistor cell of the plurality of transistor cells; the second transistor cell is another transistor cell of the plurality of transistor cells; the plurality of transistor cells comprises a third transistor cell; the plurality of transistor cells comprises a fourth transistor cell; each of the first transistor cell, the second transistor cell, the third transistor cell, and the fourth transistor cell are disposed in a first column of the plurality of columns; the third transistor cell is spaced further from the fourth transistor cell than any other transistor cells of the plurality of transistor cells that are disposed in the first column; and both the first readout circuit and the second readout circuit are disposed laterally between the third transistor cell and the fourth transistor cell.

In some embodiments, the present application provides an image sensor. The image sensor comprises a first chip. The

first chip comprises a first semiconductor substrate. The first chip comprises a first transistor cell and a second transistor cell that are disposed in a transistor cell array, wherein the first transistor cell comprises a first plurality of transistors disposed on the first semiconductor substrate, wherein the first transistor cell is configured to operate a first photodetector cell, wherein the second transistor cell comprises a second plurality of transistors disposed on the first semiconductor substrate, wherein the second transistor cell is configured to operate a second photodetector cell, and wherein a first region of the first chip laterally separates the second transistor cell from the first transistor cell. The first chip comprises a first readout circuit that is electrically coupled to the first transistor cell, wherein the first transistor cell is configured to provide a first signal to the first readout circuit that corresponds to a number of charges accumulated in a photodetector of the first photodetector cell, and wherein the first readout circuit is disposed within a perimeter of the first region of the first chip. The first chip comprises a first interlayer dielectric (ILD) structure that is disposed along a side of the first semiconductor substrate. The first chip comprises a first through-substrate via (TSV) that is disposed in the first semiconductor substrate and, at least partially, in the first ILD structure, wherein the first TSV extends vertically through the first semiconductor substrate, wherein the first TSV is electrically coupled to the first readout circuit, and wherein the first readout circuit is configured to provide a second signal to the first TSV that is based on the first signal. The image sensor comprises a second chip. The second chip comprises a second semiconductor substrate, wherein the second chip is bonded to the first chip. The second chip comprises a second ILD structure, wherein the second semiconductor substrate is disposed vertically between the first ILD structure and the second ILD structure. The second chip comprises a second readout circuit electrically coupled to the first TSV, wherein the second readout circuit is configured to receive the second signal and output a third signal that is based on the second signal.

In some embodiments, the first transistor cell comprises a first isolation structure that is disposed in the first semiconductor substrate; the second transistor cell comprises a second isolation structure that is disposed in the first semiconductor substrate; a sidewall of the first isolation structure faces a sidewall of the second isolation structure; and the sidewall of the first isolation structure and the sidewall of the second isolation structure define, at least partially, the perimeter of the first region of the first chip.

In some embodiments, the first chip comprises a third readout circuit electrically coupled to the second transistor cell; the second transistor cell is configured to provide a fourth signal to the third readout circuit that corresponds to a number of charges accumulated in a photodetector of the second photodetector cell; the third readout circuit is disposed within a perimeter of a second region of the first chip; the second transistor cell laterally separates the second region of the first chip from the first region of the first chip; the first chip comprises a second TSV disposed in the first semiconductor substrate and, at least partially, in the first ILD structure; the second TSV extends vertically through the first semiconductor substrate; the second TSV is disposed within the perimeter of the second region of the first chip; the second TSV is electrically coupled to the third readout circuit; the second readout circuit is electrically coupled to the second TSV; the third readout circuit is configured to provide a fifth signal to the second TSV that is based on the fourth signal; and the second readout circuit

is configured to receive the fifth signal and output a sixth signal that is based on the fifth signal.

In some embodiments, the second chip comprises a conductive wire disposed in the second ILD structure; a first portion of the conductive wire is disposed between the first TSV and the second readout circuit; the first TSV is electrically coupled to the second readout circuit via the first portion of the conductive wire; a second portion of the conductive wire is disposed between the second TSV and the second readout circuit; and the second TSV is electrically coupled to the second readout circuit via the second portion of the conductive wire.

In some embodiments, the image sensor further comprises a third chip. The third chip comprises a third semiconductor substrate, wherein the third chip is bonded to the first chip, wherein the photodetector of the first photodetector cell is disposed in the third semiconductor substrate, and wherein the first chip is disposed vertically between the second chip and the third chip.

In some embodiments, the present application provides an image sensor. The image sensor includes a first chip having a plurality of transistor cells disposed within rows and columns of a transistor cell array, the plurality of transistor cells respectively including a plurality of transistors configured to operate a photodetector cell; a second chip having a plurality of readout circuits; and a plurality of through-substrate vias (TSVs) extending through the first chip, wherein the plurality of TSVs including a first group of multiple TSVs that are arranged within a first row of the transistor cell array and that are configured to electrically couple a first group of the plurality of transistor cells within the first row to one or more first readout circuits of the plurality of readout circuits; and a second group of multiple TSVs that are arranged within a second row of the transistor cell array that are configured to electrically couple a second group of the plurality of transistor cells within the second row to one or more second readout circuits of the plurality of readout circuits.

In some embodiments, the plurality of transistor cells include a first transistor cell having a first isolation structure that is disposed in a first semiconductor substrate; a second transistor cell having a second isolation structure that is disposed in the first semiconductor substrate; a sidewall of the first isolation structure facing a sidewall of the second isolation structure; and the sidewall of the first isolation structure and the sidewall of the second isolation structure defining, at least partially, a perimeter of a first region of the first chip, at least one of the plurality of readout circuits disposed within the perimeter of the first region of the first chip.

In some embodiments, each of the plurality of transistor cells is coupled to a separate readout circuit of the plurality of readout circuits by way of a separate TSV of the plurality of TSVs.

In some embodiments a number of the plurality of readout circuits in the second chip is equal to a number of the plurality of TSVs.

In some embodiments, the image sensor further includes a third chip having a third semiconductor substrate, the third chip being bonded to the first chip, a photodetector of a first photodetector cell being disposed in the third semiconductor substrate, and the first chip being disposed vertically between the second chip and the third chip.

In some embodiments, the present application provides a method for forming an image sensor. The method comprises forming a first chip. Forming the first chip comprises receiving a workpiece. The workpiece comprises: a first transistor

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cell disposed on a first semiconductor substrate; a second transistor cell disposed on the first semiconductor substrate, wherein the second transistor cell is laterally spaced from the first transistor cell; a first interlayer dielectric (ILD) structure disposed along a first side of the first semiconductor substrate; and a conductive wire disposed in the first ILD structure, wherein the first transistor cell and the second transistor cell are both electrically coupled to the conductive wire. An opening is formed in the first semiconductor substrate and in the first ILD structure, wherein the opening exposes a portion of the conductive wire that is disposed laterally between the first transistor cell and the second transistor cell. A through-substrate via (TSV) is formed in the opening, wherein the TSV is formed electrically coupled to the conductive wire. A second chip is bonded to the first chip. The second chip comprises a second semiconductor substrate and a second ILD structure. The second chip comprises a readout circuit comprising an amplifier circuit. The second chip comprises a first bond pad that is electrically coupled to the amplifier circuit. The second chip is bonded to the first chip so that the first semiconductor substrate is vertically between the second ILD structure and the first ILD structure. The second chip is bonded to the first chip so that the TSV is electrically coupled to the first bond pad.

In some embodiments, forming the first chip further comprises forming a dielectric structure along a second side of the first semiconductor substrate, wherein the second side of the first semiconductor substrate is opposite the first side of the first semiconductor substrate. Forming the first chip further comprises forming a second bond pad in the dielectric structure. The second chip is bonded to the first chip so that the first bond pad is bonded to the second bond pad.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An image sensor comprising:

a first chip comprising:

a first semiconductor substrate, wherein the first semiconductor substrate has a first side and a second side, and wherein the second side of the first semiconductor substrate is opposite the first side of the first semiconductor substrate;

a first transistor cell and a second transistor cell disposed in a transistor cell array, wherein the first transistor cell comprises a first plurality of transistors disposed along the first side of the first semiconductor substrate, wherein the first transistor cell is configured to operate a first photodetector cell, wherein the second transistor cell comprises a second plurality of transistors disposed along the first side of the first semiconductor substrate, wherein the second transistor cell is configured to operate a second photodetector cell, and wherein the second transistor cell is laterally spaced from the first transistor cell;

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a first through-substrate via (TSV) extending vertically through the first semiconductor substrate, wherein the first transistor cell is electrically coupled to the first TSV, wherein the first transistor cell is configured to provide a first signal to the first TSV that corresponds to a number of charges accumulated in a photodetector of the first photodetector cell; and a second TSV extending vertically through the first semiconductor substrate, wherein the second transistor cell is electrically coupled to the second TSV, wherein the second transistor cell is configured to provide a second signal to the second TSV that corresponds to a number of charges accumulated in a photodetector of the second photodetector cell;

a second chip comprising:

a second semiconductor substrate, wherein the second chip is bonded to the first chip;

a first interlayer dielectric (ILD) structure disposed vertically between the second semiconductor substrate and the first semiconductor substrate, and wherein the first side of the first semiconductor substrate is disposed vertically between the second side of the first semiconductor substrate and the second semiconductor substrate; and

a first readout circuit electrically coupled to the first TSV and to the second TSV, wherein the first readout circuit is disposed laterally, at least partially, between the first TSV and the second TSV.

2. The image sensor of claim 1, wherein the first readout circuit comprises a first part that is within the second chip and a second part that is within a third chip bonded to the second chip.

3. The image sensor of claim 1, wherein the first readout circuit comprises an amplifier circuit comprising one or more first readout devices disposed on the second semiconductor substrate.

4. The image sensor of claim 3, wherein the one or more first readout devices are disposed laterally between the first TSV and the second TSV.

5. The image sensor of claim 1, wherein:

the first transistor cell comprises a first select transistor; a source/drain region of the first select transistor is electrically coupled to the first TSV;

the second transistor cell comprises a second select transistor; and

a source/drain region of the second select transistor is electrically coupled to the second TSV.

6. The image sensor of claim 1, wherein:

the first TSV is laterally spaced from the first transistor cell and the second transistor cell by at least about 0.1 micrometers; and

the second TSV is laterally spaced from the first transistor cell and the second transistor cell by at least about 0.1 micrometers.

7. The image sensor of claim 1, further comprising:

a third chip comprising:

a third semiconductor substrate, wherein the third chip is bonded to the first chip, wherein the photodetector of the first photodetector cell and the photodetector of the second photodetector cell are both disposed in the third semiconductor substrate, and wherein the first chip is disposed vertically between the second chip and the third chip.

8. The image sensor of claim 1, wherein:

the transistor cell array comprises a plurality of transistor cells;

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the transistor cell array comprises a plurality of rows and a plurality of columns;
 the first transistor cell is one transistor cell of the plurality of transistor cells;
 the second transistor cell is another transistor cell of the plurality of transistor cells; and
 both the first transistor cell and the second transistor cell are disposed in a first column of the plurality of columns.

9. The image sensor of claim 8, wherein the first transistor cell neighbors the second transistor cell.

10. The image sensor of claim 8, wherein the first transistor cell is spaced further from the second transistor cell than any other transistor cell of the plurality of transistor cells that are disposed in the first column.

11. The image sensor of claim 8, wherein:
 the first readout circuit is spaced laterally from the first transistor cell by a first distance;
 the first readout circuit is spaced laterally from the second transistor cell by a second distance; and
 the first distance is about the same as the second distance.

12. The image sensor of claim 1, wherein:
 the second chip comprises a second readout circuit electrically coupled to the second TSV;
 the second readout circuit is configured to receive the second signal; and
 the second readout circuit is laterally spaced from the first readout circuit.

13. The image sensor of claim 12, wherein:
 the transistor cell array comprises a plurality of transistor cells;
 the transistor cell array comprises a plurality of rows and a plurality of columns;
 the first transistor cell is one transistor cell of the plurality of transistor cells;
 the second transistor cell is another transistor cell of the plurality of transistor cells;
 the plurality of transistor cells comprises a third transistor cell;
 the plurality of transistor cells comprises a fourth transistor cell;
 each of the first transistor cell, the second transistor cell, the third transistor cell, and the fourth transistor cell are disposed in a first column of the plurality of columns;
 the third transistor cell is spaced further from the fourth transistor cell than any other transistor cells of the plurality of transistor cells that are disposed in the first column; and
 both the first readout circuit and the second readout circuit are disposed laterally between the third transistor cell and the fourth transistor cell.

14. An image sensor comprising:
 a first chip comprising a plurality of transistor cells disposed within rows and columns of a transistor cell array, wherein the plurality of transistor cells respectively comprise a plurality of transistors configured to operate a photodetector cell;
 a second chip comprising a plurality of readout circuits; and
 a plurality of through-substrate vias (TSVs) extending through the first chip, wherein the plurality of TSVs comprise:
 a first group of multiple TSVs that are arranged within a first row of the transistor cell array and that are configured to electrically couple a first group of the

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plurality of transistor cells within the first row to one or more first readout circuits of the plurality of readout circuits; and
 a second group of multiple TSVs that are arranged within a second row of the transistor cell array that are configured to electrically couple a second group of the plurality of transistor cells within the second row to one or more second readout circuits of the plurality of readout circuits.

15. The image sensor of claim 14, wherein the plurality of transistor cells comprise:
 a first transistor cell comprising a first isolation structure that is disposed in a first semiconductor substrate; and
 a second transistor cell comprising a second isolation structure that is disposed in the first semiconductor substrate;
 wherein a sidewall of the first isolation structure faces a sidewall of the second isolation structure; and
 wherein the sidewall of the first isolation structure and the sidewall of the second isolation structure define, at least partially, a perimeter of a first region of the first chip, at least one of the plurality of readout circuits disposed within the perimeter of the first region of the first chip.

16. The image sensor of claim 14, wherein each of the plurality of transistor cells is coupled to a separate readout circuit of the plurality of readout circuits by way of a separate TSV of the plurality of TSVs.

17. The image sensor of claim 16, wherein a number of the plurality of readout circuits in the second chip is equal to a number of the plurality of TSVs.

18. The image sensor of claim 14, further comprising:
 a third chip comprising:
 a third semiconductor substrate, wherein the third chip is bonded to the first chip, wherein a photodetector of a first photodetector cell is disposed in the third semiconductor substrate, and wherein the first chip is disposed vertically between the second chip and the third chip.

19. A method for forming an image sensor, the method comprising:
 forming a first chip, wherein forming the first chip comprises:
 receiving a workpiece, wherein the workpiece comprises:
 a first transistor cell disposed on a first semiconductor substrate;
 a second transistor cell disposed on the first semiconductor substrate,
 wherein the second transistor cell is laterally spaced from the first transistor cell;
 a first interlayer dielectric (ILD) structure disposed along a first side of the first semiconductor substrate;
 a conductive wire disposed in the first ILD structure, wherein the first transistor cell and the second transistor cell are both electrically coupled to the conductive wire;
 forming an opening in the first semiconductor substrate and in the first ILD structure, wherein the opening exposes a portion of the conductive wire that is disposed laterally between the first transistor cell and the second transistor cell; and
 forming a through-substrate via (TSV) in the opening, wherein the TSV is formed electrically coupled to the conductive wire;
 bonding a second chip to the first chip, wherein:

the second chip comprises a second semiconductor substrate and a second ILD structure;

the second chip comprises a readout circuit comprising an amplifier circuit;

the second chip comprises a first bond pad that is electrically coupled to the amplifier circuit; 5

the second chip is bonded to the first chip so that the first semiconductor substrate is vertically between the second ILD structure and the first ILD structure; and 10

the second chip is bonded to the first chip so that the TSV is electrically coupled to the first bond pad.

20. The method of claim **19**, wherein:

forming the first chip further comprises forming a dielectric structure along a second side of the first semiconductor substrate, wherein the second side of the first semiconductor substrate is opposite the first side of the first semiconductor substrate; 15

forming the first chip further comprises forming a second bond pad in the dielectric structure; and 20

the second chip is bonded to the first chip so that the first bond pad is bonded to the second bond pad.

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